Top 131 CAT Number Systems Questions With Video Solutions

Questions

Instructions

For the following questions answer them individually

Question 1

How many pairs of positive integers m, n satisfy 1/m + 4/n = 1/12 , where n is an odd integer less than 60?

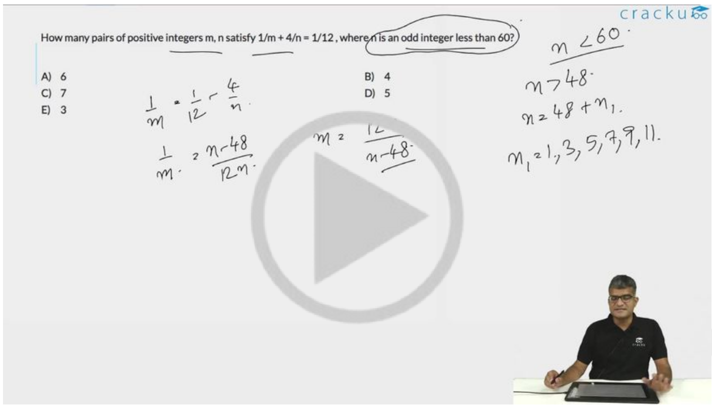
A 6

B 4

C 7

D 5

E 3



VIDEO SOLUTION

Question 2

In a tournament, there are n teams $T_1, T_2....., T_n$ with $n > 5$. Each team consists of k players, $k > 3$. The following pairs of teams have one player in common: T_1 & T_2 , T_2 & T_3 ,......, T_{n-1} & T_n , and T_n & T_1 . No other pair of teams has any player in common. How many players are participating in the tournament, considering all the n teams together?

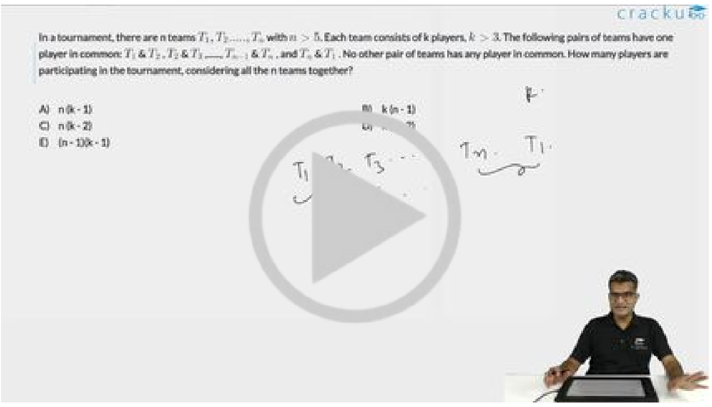
A n (k - 1)

B k (n - 1)

C n (k - 2)

D k (k - 2)

E (n - 1)(k - 1)



VIDEO SOLUTION

Question 3

Consider four digit numbers for which the first two digits are equal and the last two digits are also equal. How many such numbers are perfect squares?

A 3

B 2

C 4

D 0

E 1

VIDEO SOLUTION

Question 4

The integers 1, 2, ..., 40 are written on a blackboard. The following operation is then repeated 39 times: In each repetition, any two numbers, say a and b, currently on the blackboard are erased and a new number a + b - 1 is written. What will be the number left on the board at the end?

A 820

B 821

C 781

D 819

E 780

VIDEO SOLUTION

Question 5

What are the last two digits of 7^{2008}?

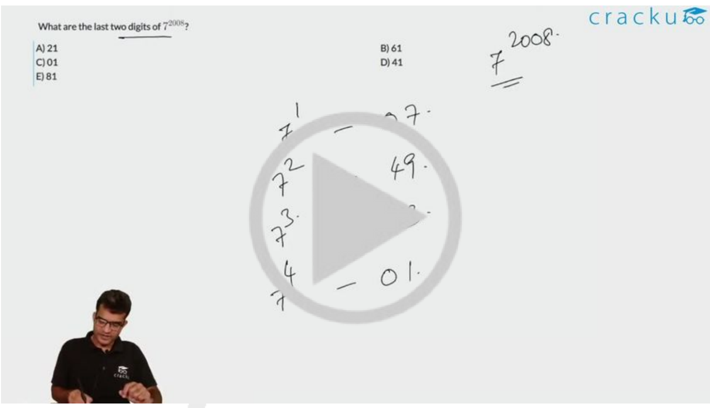
A 21

B 61

C 01

D 41

E 81



VIDEO SOLUTION

Question 6

A shop stores x kg of rice. The first customer buys half this amount plus half a kg of rice. The second customer buys half the remaining amount plus half a kg of rice. Then the third customer also buys half the remaining amount plus half a kg of rice. Thereafter, no rice is left in the shop. Which of the following best describes the value of x?

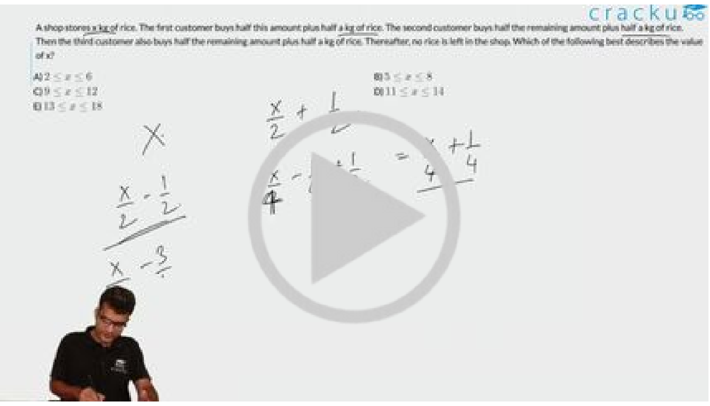
A $2 \leq x \leq 6$

B $5 \leq x \leq 8$

C $9 \leq x \leq 12$

D $11 \leq x \leq 14$

E $13 \leq x \leq 18$



VIDEO SOLUTION

Question 7

Three consecutive positive integers are raised to the first, second and third powers respectively and then added. The sum so obtained is perfect square whose square root equals the total of the three original integers. Which of the following best describes the minimum, say m, of these three integers?

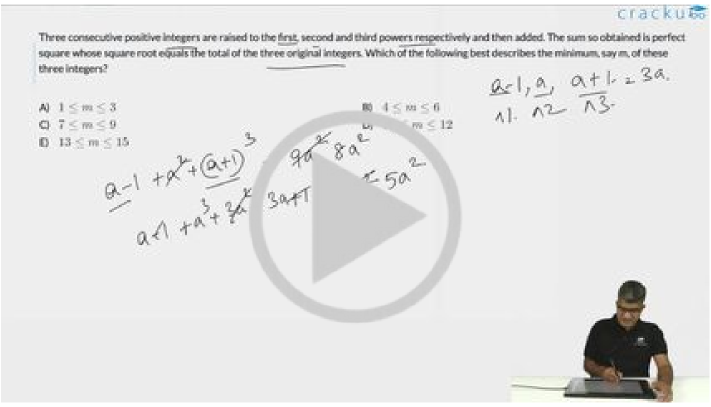
A $1 \leq m \leq 3$

B $4 \leq m \leq 6$

C $7 \leq m \leq 9$

D $10 \leq m \leq 12$

E $13 \leq m \leq 15$



VIDEO SOLUTION

Question 8

Twenty-seven persons attend a party. Which one of the following statements can never be true?

A There is a person in the party who is acquainted with all the twenty-six others.

B Each person in the party has a different number of acquaintances.

C There is a person in the party who has an odd number of acquaintances.

D In the party, there is no set of three mutual acquaintances.

VIEW SOLUTION

Question 9

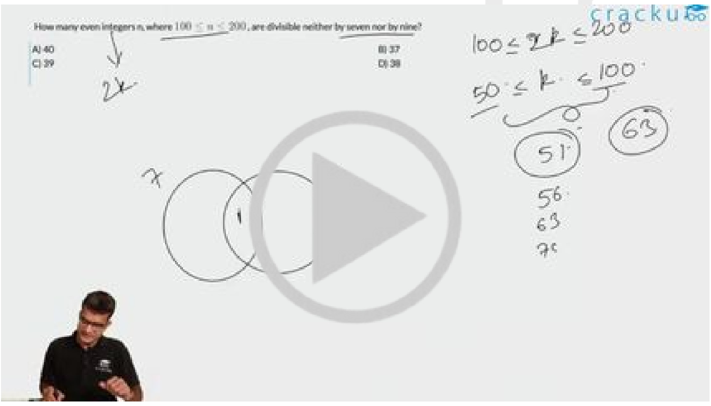
How many even integers n, where $100 \leq n \leq 200$, are divisible neither by seven nor by nine?

A 40

B 37

C 39

D 38



VIDEO SOLUTION

Question 10

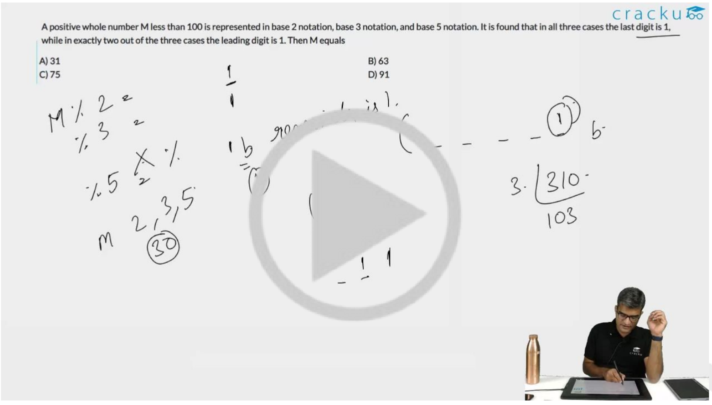
A positive whole number M less than 100 is represented in base 2 notation, base 3 notation, and base 5 notation. It is found that in all three cases the last digit is 1, while in exactly two out of the three cases the leading digit is 1. Then M equals

A 31

B 63

C 75

D 91



VIDEO SOLUTION

Question 11

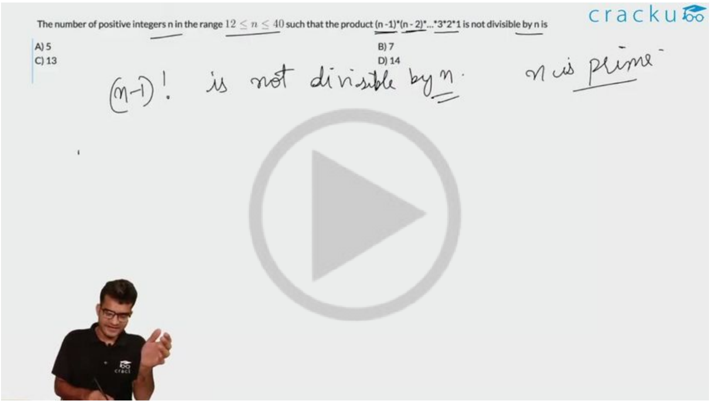
The number of positive integers n in the range $12 \leq n \leq 40$ such that the product (n -1)*(n - 2)*...*3*2*1 is not divisible by n is

A 5

B 7

C 13

D 14



VIDEO SOLUTION

Question 12

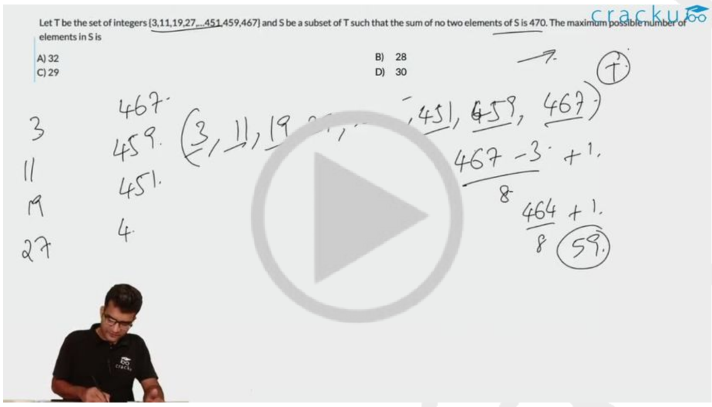
Let T be the set of integers {3,11,19,27,...451,459,467} and S be a subset of T such that the sum of no two elements of S is 470. The maximum possible number of elements in S is

A 32

B 28

C 29

D 30



VIDEO SOLUTION

Question 13

Suppose n is an integer such that the sum of digits on n is 2, and $10^{10} < n < 10^{11}$. The number of different values of n is

A 11

B 10

C 9

D 8

VIEW SOLUTION

Question 14

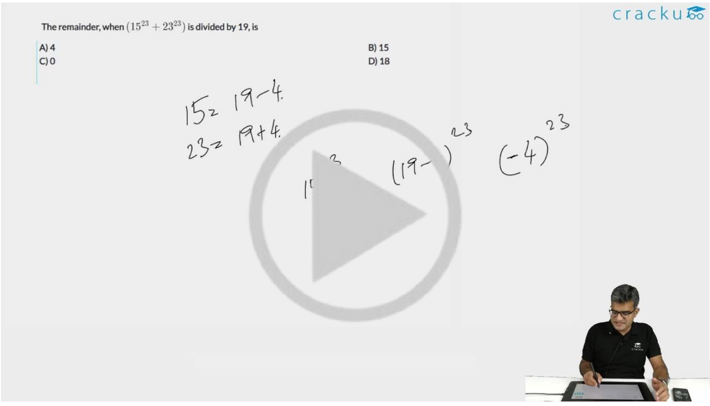
The remainder, when $(15^{23} + 23^{23})$ is divided by 19, is

A 4

B 15

C 0

D 18



VIDEO SOLUTION

Question 15

If a/b = 1/3, b/c = 2, c/d = 1/2 , d/e = 3 and e/f = 1/4, then what is the value of abc/def ?

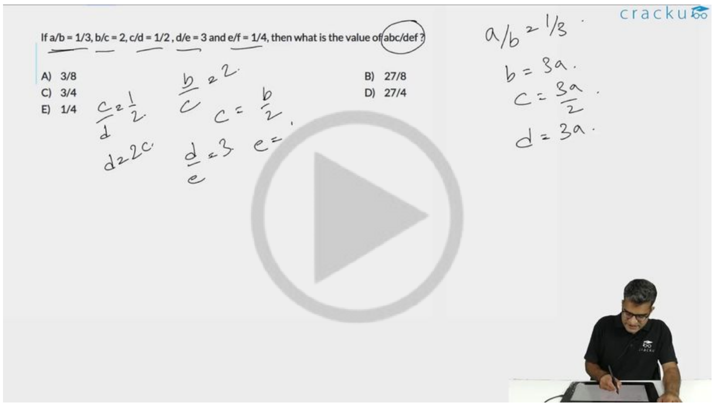
A 3/8

B 27/8

C 3/4

D 27/4

E 1/4



VIDEO SOLUTION

Question 16

The sum of four consecutive two-digit odd numbers, when divided by 10, becomes a perfect square. Which of the following can possibly be one of these four numbers?

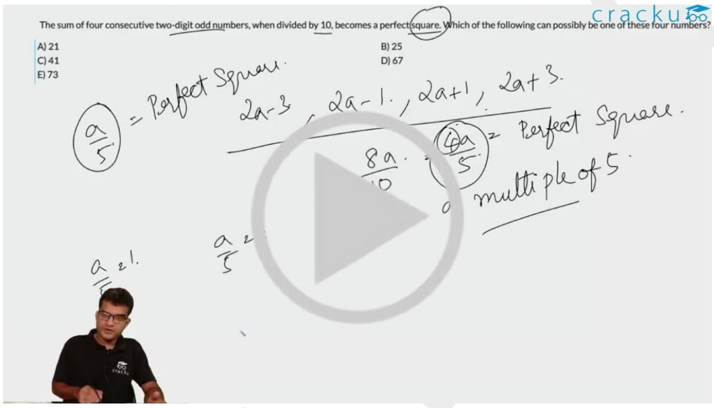
A 21

B 25

C 41

D 67

E 73



VIDEO SOLUTION

Question 17

The number of employees in Obelix Menhir Co. is a prime number and is less than 300. The ratio of the number of employees who are graduates and above, to that of employees who are not, can possibly be:

A 101:88

B 87:100

C 110:111

D 85:98

E 97:84

VIEW SOLUTION

Question 18

If R = $(30^{65} - 29^{65})/(30^{64} + 29^{64})$,then

A $0 < R \leq 0.1$

B $0.1 < R \leq 0.5$

C $0.5 < R \leq 1.0$

D $R > 1.0$

VIEW SOLUTION

Question 19

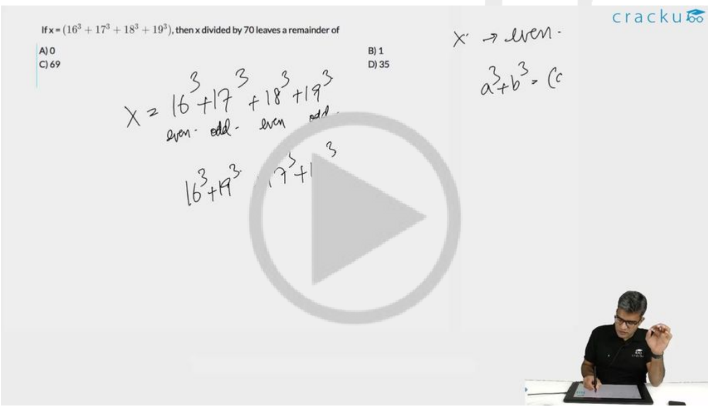
If x = $(16^3 + 17^3 + 18^3 + 19^3)$, then x divided by 70 leaves a remainder of

A 0

B 1

C 69

D 35



VIDEO SOLUTION

Question 20

Let $n! = 1 * 2 * 3 * ... * n$ for integer $n \geq 1$.

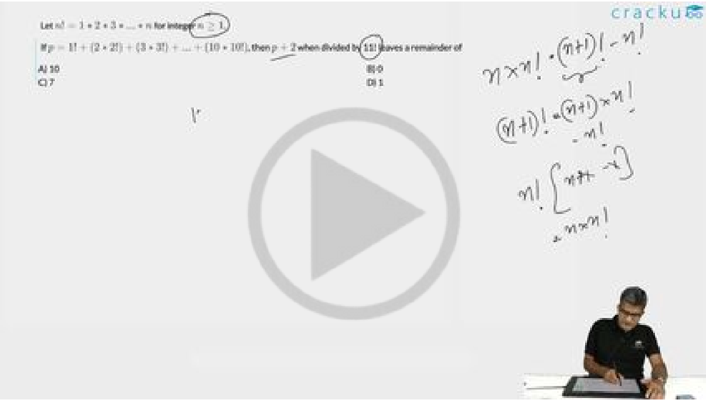
If $p = 1! + (2 * 2!) + (3 * 3!) + ... + (10 * 10!)$, then $p + 2$ when divided by 11! leaves a remainder of

A 10

B 0

C 7

D 1



VIDEO SOLUTION

Question 21

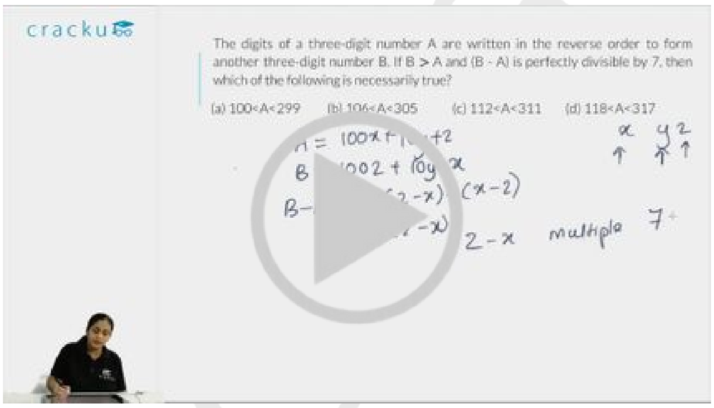
The digits of a three-digit number A are written in the reverse order to form another three-digit number B. If B > A and B-A is perfectly divisible by 7, then which of the following is necessarily true?

A 100<A<299

B 106<A<305

C 112<A<311

D 118<A<317



VIDEO SOLUTION

Question 22

The rightmost non-zero digit of the number 30^{2720} is

A 1

B 3

C 7

D 9

VIDEO SOLUTION

Question 23

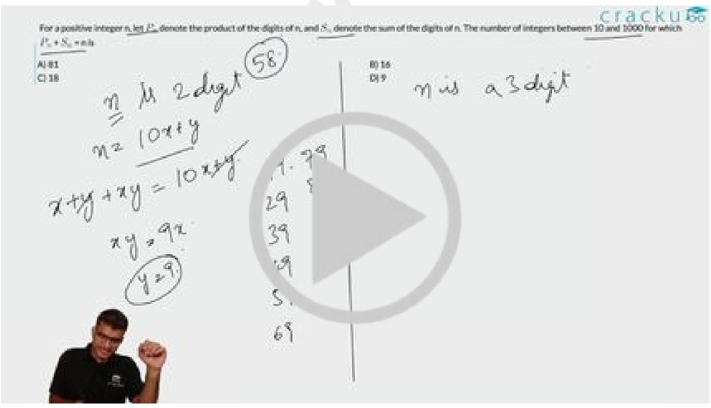
For a positive integer n, let P_n denote the product of the digits of n, and S_n denote the sum of the digits of n. The number of integers between 10 and 1000 for which P_n + S_n = n is

A 81

B 16

C 18

D 9



VIDEO SOLUTION

Question 24

Let S be a set of positive integers such that every element n of S satisfies the conditions

A. 1000 <= n <= 1200

B. every digit in n is odd

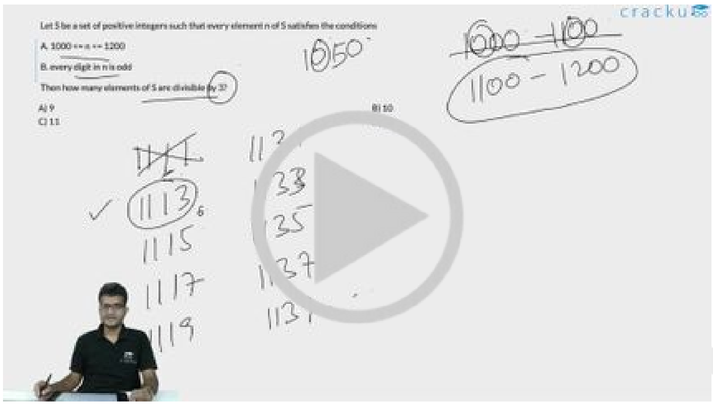
Then how many elements of S are divisible by 3?

A 9

B 10

C 11

D 12



VIDEO SOLUTION

Question 25

Let x, y and z be distinct integers. x and y are odd and positive, and z is even and positive. Which one of the following statements cannot be true?

A $y(x-z)^2$ is even

B $y^2(x-z)$ is odd

C $y(x-z)$ is odd

D $z(x-y)^2$ is even

VIEW SOLUTION

Question 26

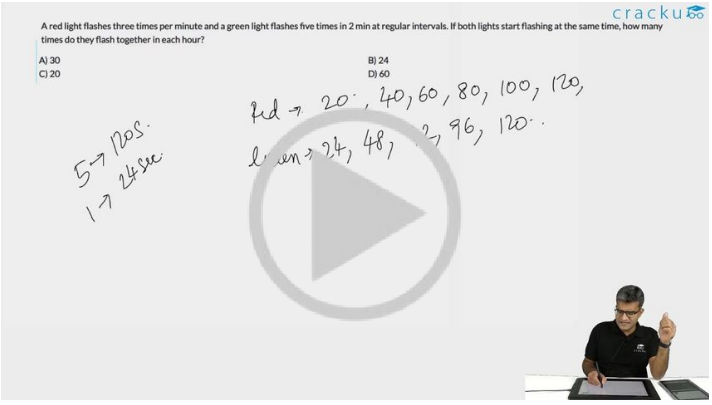
A red light flashes three times per minute and a green light flashes five times in 2 min at regular intervals. If both lights start flashing at the same time, how many times do they flash together in each hour?

A 30

B 24

C 20

D 60



VIDEO SOLUTION

Question 27

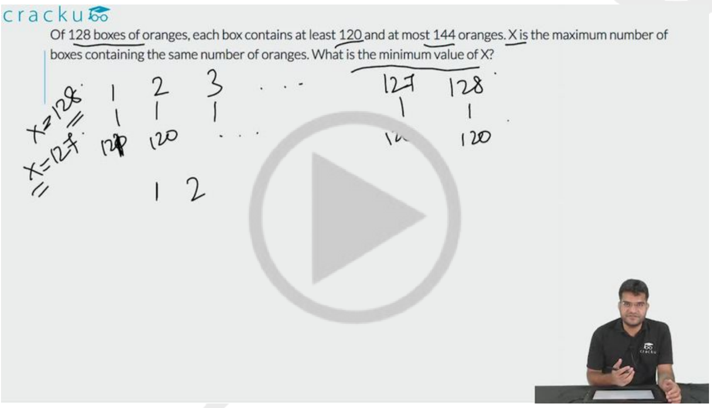
Of 128 boxes of oranges, each box contains at least 120 and at most 144 oranges. X is the maximum number of boxes containing the same number of oranges. What is the minimum value of X?

A 5

B 103

C 6

D Cannot be determined



VIDEO SOLUTION

Question 28

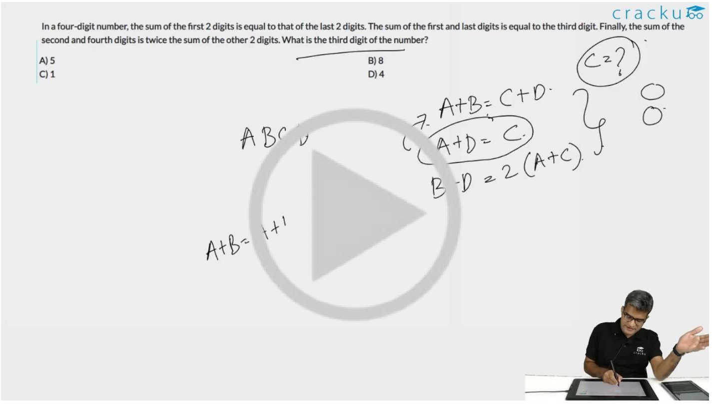
In a four-digit number, the sum of the first 2 digits is equal to that of the last 2 digits. The sum of the first and last digits is equal to the third digit. Finally, the sum of the second and fourth digits is twice the sum of the other 2 digits. What is the third digit of the number?

A 5

B 8

C 1

D 4



VIDEO SOLUTION

Question 29

Anita had to do a multiplication. In stead of taking 35 as one of the multipliers, she took 53. As a result, the product went up by 540. What is the actual product?

A 1050

B 540

C 1440

D 1590

VIEW SOLUTION

Question 30

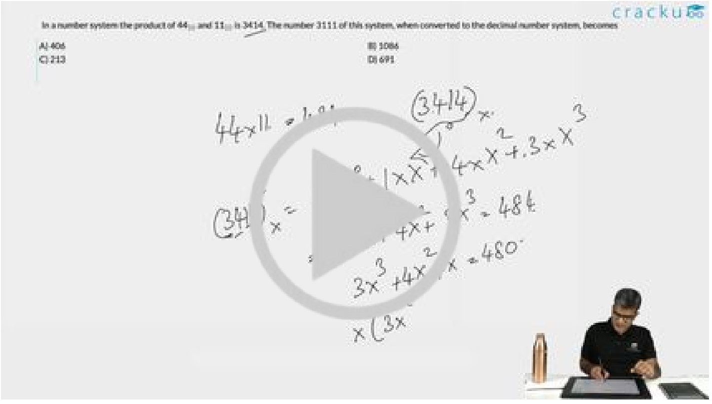
In a number system the product of 44_{10} and 11_{10} is 3414. The number 3111 of this system, when converted to the decimal number system, becomes

A 406

B 1086

C 213

D 691



VIDEO SOLUTION

Question 31

All the page numbers from a book are added, beginning at page 1. However, one page number was added twice by mistake. The sum obtained was 1000. Which page number was added twice?

A 44

B 45

C 10

D 12

VIEW SOLUTION

Question 32

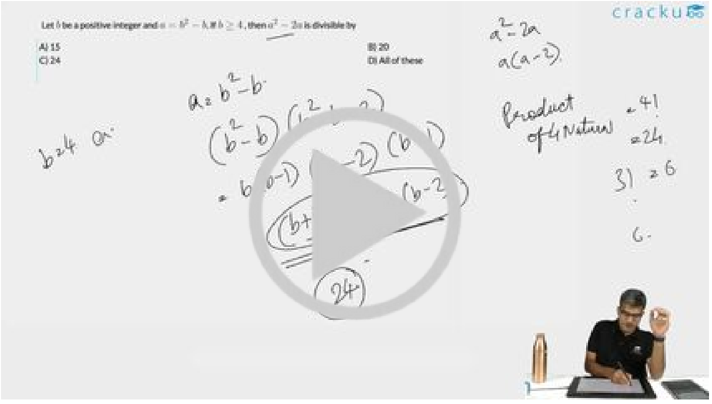
Let b be a positive integer and $a = b^2 - b$. If $b \geq 4$, then $a^2 - 2a$ is divisible by

A 15

B 20

C 24

D All of these



VIDEO SOLUTION

Question 33

Ashish is given Rs. 158 in one-rupee denominations. He has been asked to allocate them into a number of bags such that any amount required between Re 1 and Rs. 158 can be given by handing out a certain number of bags without opening them. What is the minimum number of bags required?

A 11

B 12

C 13

D None of these

VIEW SOLUTION

Question 34

In some code, letters a, b, c, d and e represent numbers 2, 4, 5, 6 and 10. We just do not know which letter represents which number. Consider the following relationships:

I. a + c = e,
II. b - d = d and
III. e + a = b

Which of the following statements is true?

A b = 4, d = 2

B a = 4, e = 6

C b = 6, e = 2

D a = 4, c = 6

VIEW SOLUTION

Question 35

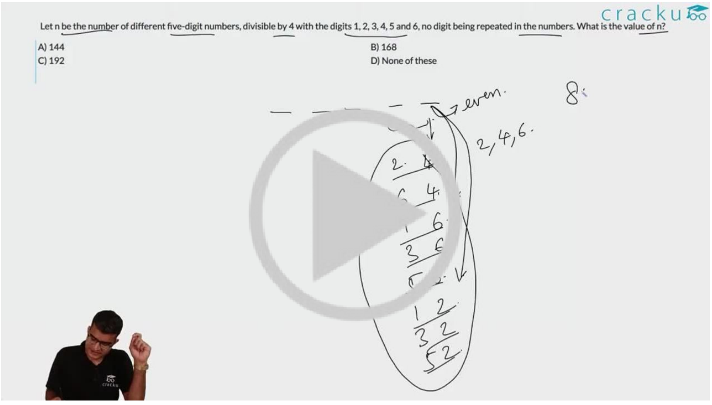
Let n be the number of different five-digit numbers, divisible by 4 with the digits 1, 2, 3, 4, 5 and 6, no digit being repeated in the numbers. What is the value of n?

A 144

B 168

C 192

D None of these



VIDEO SOLUTION

Question 36

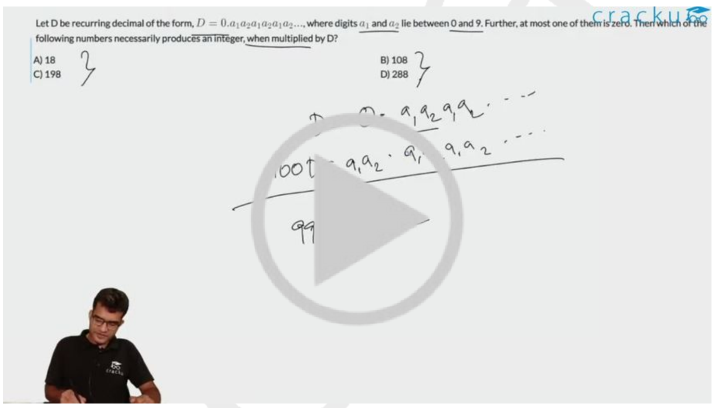
Let D be recurring decimal of the form, $D = 0.a_1a_2a_1a_2a_1a_2...$, where digits a_1 and a_2 lie between 0 and 9. Further, at most one of them is zero. Then which of the following numbers necessarily produces an integer, when multiplied by D?

A 18

B 108

C 198

D 288



VIDEO SOLUTION

Question 37

Let S be the set of integers x such that:

1) 100 <= x <= 200

2) x is odd

3) x is divisible by 3 but not by 7.

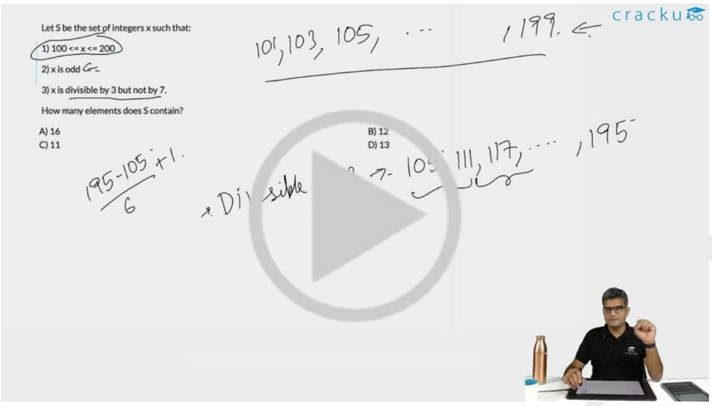
How many elements does S contain?

A 16

B 12

C 11

D 13



VIDEO SOLUTION

Question 38

Let S be the set of prime numbers greater than or equal to 2 and less than 100. Multiply all the elements of S. With how many consecutive zeroes will the product end?

A 1

B 4

C 5

D 10

VIEW SOLUTION

Question 39

Let x, y and z be distinct integers, that are odd and positive. Which one of the following statements cannot be true?

A xyz^2 is odd

B $(x-y)^2 z$ is even

C $(x+y-z)^2(x+y)$ is even

D $(x-y)(y+z)(x+y-z)$ is odd

VIEW SOLUTION

Question 40

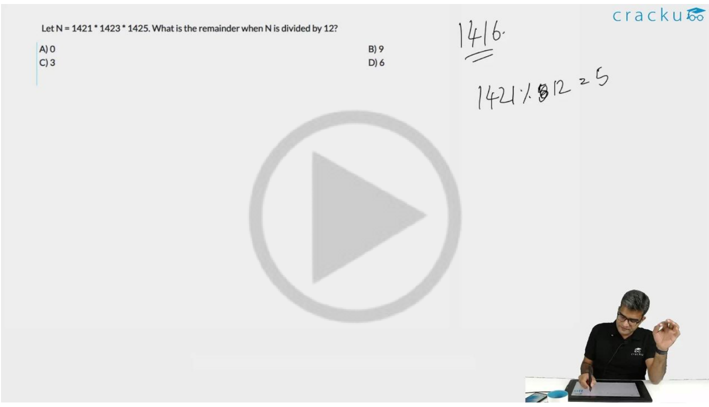
Let N = 1421 * 1423 * 1425. What is the remainder when N is divided by 12?

A 0

B 9

C 3

D 6



VIDEO SOLUTION

Question 41

The integers 34041 and 32506 when divided by a three-digit integer n leave the same remainder. What is n?

A 289

B 367

C 453

D 307

VIEW SOLUTION

Question 42

Each of the numbers $x_1, x_2, ..., x_n$ $(n > 4)$, is equal to 1 or -1.
Suppose, $x_1x_2x_3x_4 + x_2x_3x_4x_5 + x_3x_4x_5x_6 + ... + x_{n-3}x_{n-2}x_{n-1}x_nx_1 + x_{n-1}x_nx_1x_2 + x_nx_1x_2x_3$ = 0, then:

A n is even

B n is odd

C n is an odd multiple of 3

D n is prime

VIEW SOLUTION

Question 43

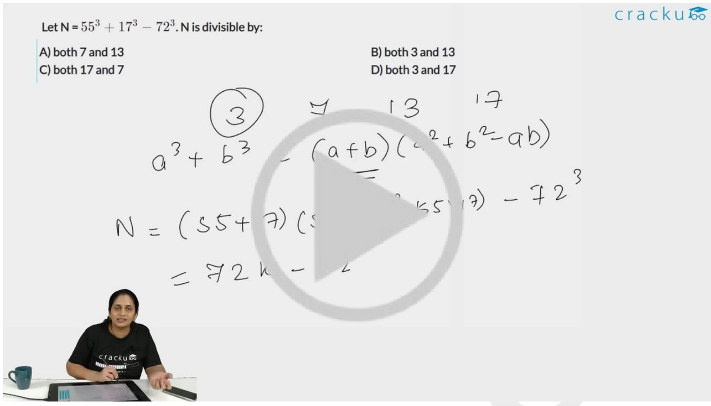
Let N = $55^3 + 17^3 - 72^3$. N is divisible by:

A both 7 and 13

B both 3 and 13

C both 17 and 7

D both 3 and 17



VIDEO SOLUTION

Question 44

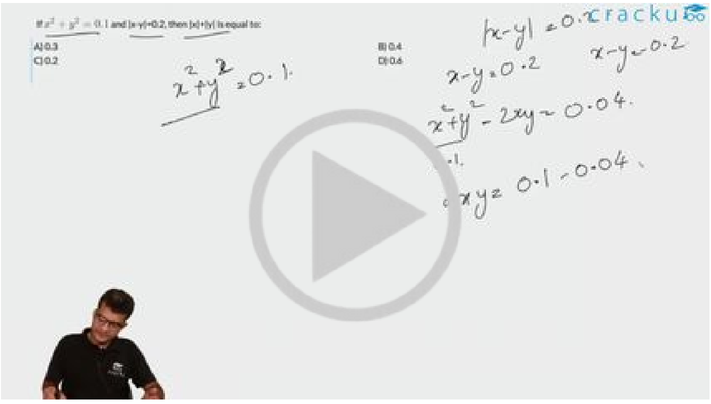
If $x^2 + y^2 = 0.1$ and |x-y|=0.2, then |x|+|y| is equal to:

A 0.3

B 0.4

C 0.2

D 0.6



VIDEO SOLUTION

Question 45

Convert the number 1982 from base 10 to base 12. The result is:

A 1182

B 1912

C 1192

D 1292

VIEW SOLUTION

Question 46

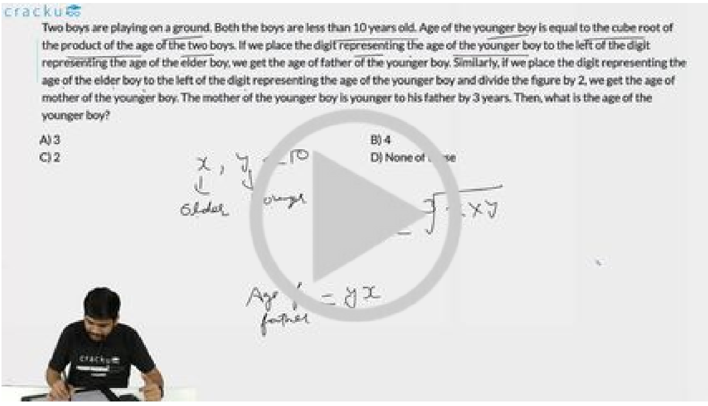
Two boys are playing on a ground. Both the boys are less than 10 years old. Age of the younger boy is equal to the cube root of the product of the age of the two boys. If we place the digit representing the age of the younger boy to the left of the digit representing the age of the elder boy, we get the age of father of the younger boy. Similarly, if we place the digit representing the age of the elder boy to the left of the digit representing the age of the younger boy and divide the figure by 2, we get the age of mother of the younger boy. The mother of the younger boy is younger to his father by 3 years. Then, what is the age of the younger boy?

A 3

B 4

C 2

D None of these



VIDEO SOLUTION

Question 47

Number S is obtained by squaring the sum of digits of a two-digit number D. If difference between S and D is 27, then the two-digit number D is

A 24

B 54

C 34

D 45

VIDEO SOLUTION

Question 48

A rich merchant had collected many gold coins. He did not want anybody to know about him. One day, his wife asked, " How many gold coins do we have?" After a brief pause, he replied, "Well! if I divide the coins into two unequal numbers, then 48 times the difference between the two numbers equals the difference between the squares of the two numbers." The wife looked puzzled. Can you help the merchant's wife by finding out how many gold coins the merchant has?

A 96

B 53

C 43

D None of these

VIEW SOLUTION

Question 49

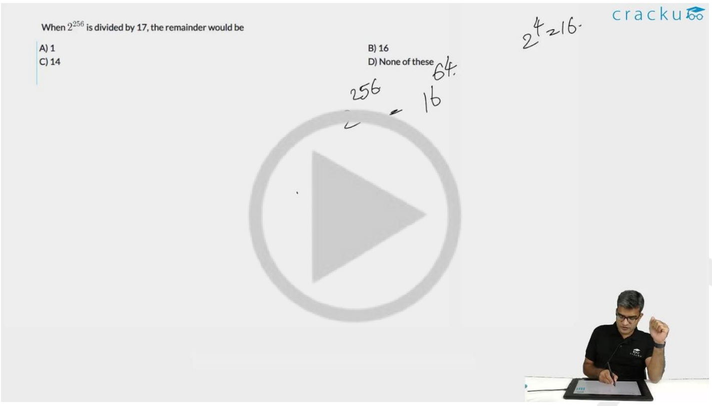
When 2^{256} is divided by 17, the remainder would be

A 1

B 16

C 14

D None of these



VIDEO SOLUTION

Question 50

At a bookstore, 'MODERN BOOK STORE' is flashed using neon lights. The words are individually flashed at the intervals of 2.5 s, 4.25 s and 5.125 s respectively, and each word is put off after a second. The least time after which the full name of the bookstore can be read again for a full second is

A 49.5 s

B 72.5 s

C 1744.5 s

D 855 s

VIEW SOLUTION

Question 51

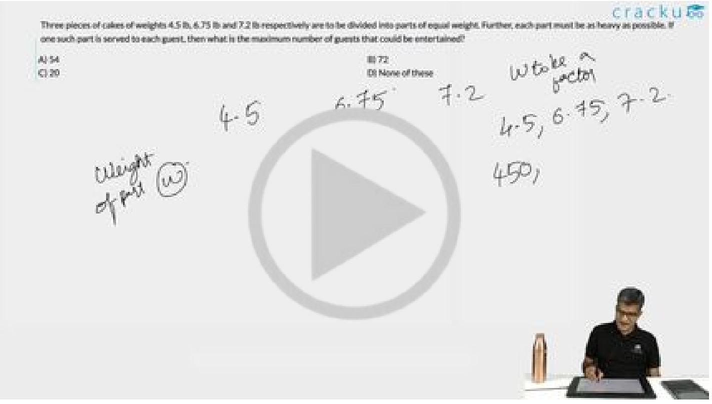
Three pieces of cakes of weights 4.5 lb, 6.75 lb and 7.2 lb respectively are to be divided into parts of equal weight. Further, each part must be as heavy as possible. If one such part is served to each guest, then what is the maximum number of guests that could be entertained?

A 54

B 72

C 20

D None of these



VIDEO SOLUTION

Question 52

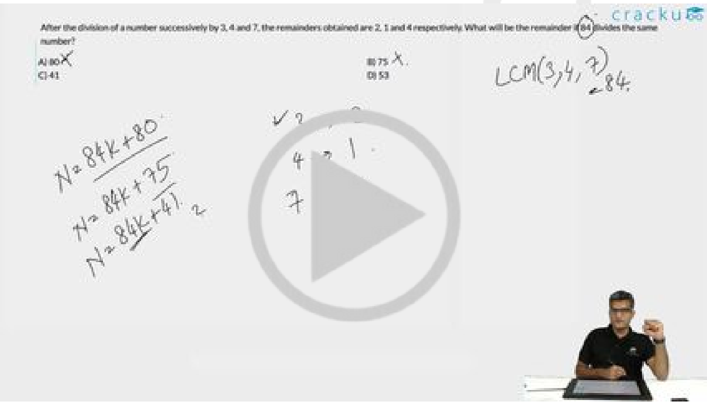
After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?

A 80

B 75

C 41

D 53



VIDEO SOLUTION

Question 53

$7^{6n} - 6^{6n}$, where n is an integer > 0, is divisible by

A 13

B 127

C 559

D All of these

VIEW SOLUTION

Question 54

The remainder when 2^{60} is divided by 5 equals

A 0

B 1

C 2

D None of these

VIEW SOLUTION

Question 55

Mr X enters a positive integer Y(>1) in an electronic calculator and then goes on pressing the square root key repeatedly. Then

A The display does not stabilize

B The display becomes closer to 0

C The display becomes closer to 1

D May not be true and the answer depends on the choice of Y

VIEW SOLUTION

Question 56

Let a, b be any positive integers and x = 0 or 1, then

A $a^x b^{1-x} = xa + (1-x)b$

B $a^x b^{1-x} = (1-x)a + xb$

C $a^x b^{(1-x)} = a(1-x)bx$

D None of the above is necessarily true.

VIEW SOLUTION

Question 57

If n is any positive integer, then $n^3 - n$ is divisible

A Always by 12

B Never by 12

C Always by 6

D Never by 6

VIEW SOLUTION

Question 58

What is the greatest power of 5 which can divide 80! exactly?

A 16

B 20

C 19

D None of these

VIDEO SOLUTION

Question 59

A third standard teacher gave a simple multiplication exercise to the kids. But one kid reversed the digits of both the numbers and carried out the multiplication and found that the product was exactly the same as the one expected by the teacher. Only one of the following pairs of numbers will fit in the description of the exercise. Which one is that?

A 14, 22

B 13, 62

C 19, 33

D 42, 28

VIEW SOLUTION

Question 60

Find the minimum integral value of n such that the division $\frac{55n}{124}$ leaves no remainder.

A 124

B 123

C 31

D 62

VIEW SOLUTION

Question 61

Let k be a positive integer such that k+4 is divisible by 7. Then the smallest positive integer n, greater than 2, such that k+2n is divisible by 7 equals

A 9

B 7

C 5

D 3

VIDEO SOLUTION

Question 62

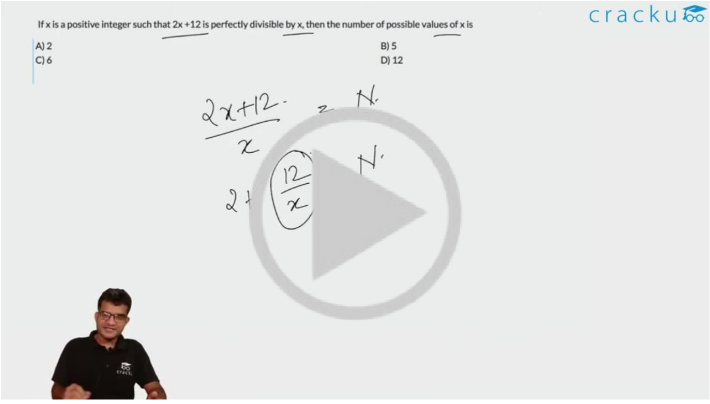
If x is a positive integer such that 2x +12 is perfectly divisible by x, then the number of possible values of x is

A 2

B 5

C 6

D 12



VIDEO SOLUTION

Question 63

A positive integer is said to be a prime number if it is not divisible by any positive integer other than itself and 1. Let p be a prime number greater than 5. Then $(p^2 - 1)$ is

A never divisible by 6

B always divisible by 6, and may or may not be divisible by 12.

C always divisible by 12, and may or may not be divisible by 24.

D always divisible by 24.

VIEW SOLUTION

Question 64

To decide whether a number of n digits is divisible by 7, we can define a process by which its magnitude is reduced as follows: $(i_1, i_2, i_3$,..... are the digits of the number, starting from the most significant digit).
$i_1 i_2 ... i_n => i_1.3^{n-1} + i_2.3^{n-2} + ... + i_n.3^0$.
e.g. $259 => 2.3^2 + 5.3^1 + 9.3^0 = 18 + 15 + 9 = 42$
Ultimately the resulting number will be seven after repeating the above process a certain number of times. After how many such stages, does the number 203 reduce to 7?

A 2

B 3

C 4

D 1

VIEW SOLUTION

Question 65

If 8 + 12 = 2, 7 + 14 = 3 then 10 + 18 = ?

A 10

B 4

C 6

D 18

VIEW SOLUTION

Question 66

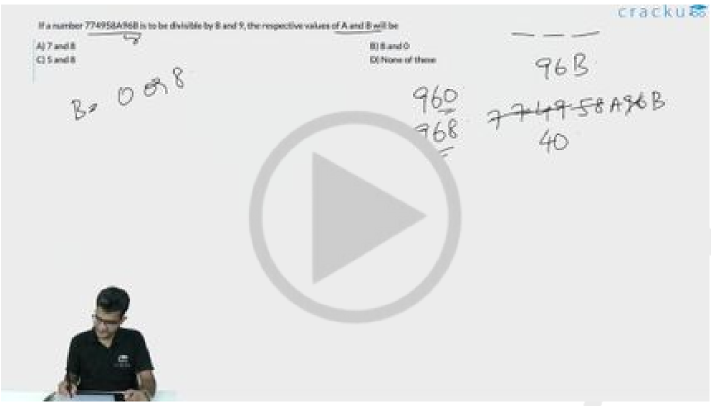
If a number 774958A96B is to be divisible by 8 and 9, the respective values of A and B will be

A 7 and 8

B 8 and 0

C 5 and 8

D None of these



VIDEO SOLUTION

Question 67

If n is any odd number greater than 1, then $n(n^2 - 1)$ is

A divisible by 96 always

B divisible by 48 always

C divisible by 24 always

D None of these

VIEW SOLUTION

Question 68

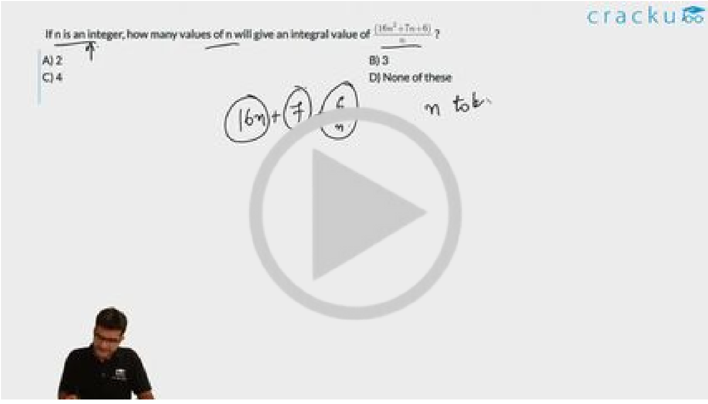
If n is an integer, how many values of n will give an integral value of $\frac{(16n^2+7n+6)}{n}$?

A 2

B 3

C 4

D None of these



VIDEO SOLUTION

Question 69

A student instead of finding the value of 7/8 of a number, found the value of 7/18 of the number. If his answer differed from the actual one by 770, find the number.

A 1584

B 2520

C 1728

D 1656

VIEW SOLUTION

Question 70

P and Q are two positive integers such that PQ = 64. Which of the following cannot be the value of P + Q?

A 20

B 65

C 16

D 35

VIEW SOLUTION

Question 71

If m and n are integers divisible by 5, which of the following is not necessarily true?

A m - n is divisible by 5

B m^2 - n^2 is divisible by 25

C m + n is divisible by 10

D None of these

VIEW SOLUTION

Question 72

P, Q and R are three consecutive odd numbers in ascending order. If the value of three times P is 3 less than two times R, find the value of R.

A 5

B 7

C 9

D 11

VIEW SOLUTION

Question 73

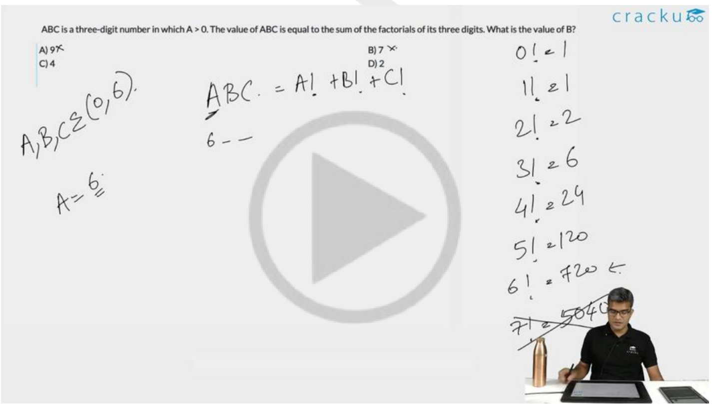
ABC is a three-digit number in which A > 0. The value of ABC is equal to the sum of the factorials of its three digits. What is the value of B?

A 9

B 7

C 4

D 2



VIDEO SOLUTION

Question 74

A, B and C are defined as follows.

A=$(2.000004) \div ((2.000004)^2 + 4.000008)$;

B = $(3.000003) \div ((3.000003)^2 + 9.000009)$

C= $(4.000002) \div ((4.000002)^2 + 8.000004)$

Which of the following is true about the values of the above three expressions?

A All of them lie between 0.18 and 0.2

B A is twice of C

C C is the smallest

D B is the smallest

VIEW SOLUTION

Question 75

n^3 is odd. Which of the following statement(s) is/are true?
I. n is odd.
II.n^2 is odd.
III.n^2 is even.

A I only

B II only

C I and II

D I and III

VIDEO SOLUTION

Question 76

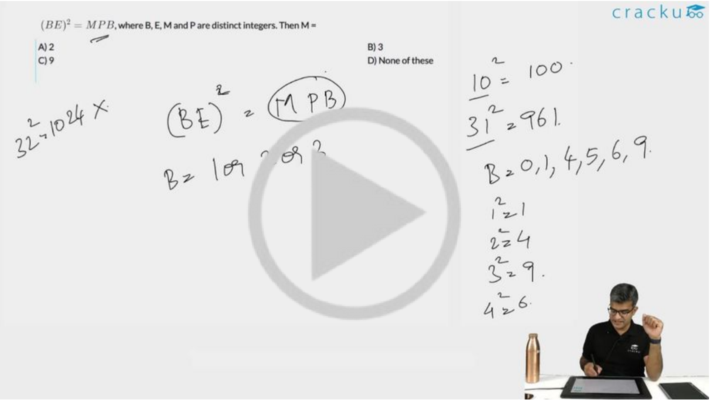
$(BE)^2 = MPB$, where B, E, M and P are distinct integers. Then M =

A 2

B 3

C 9

D None of these



VIDEO SOLUTION

Question 77

Five-digit numbers are formed using only 0, 1, 2, 3, 4 exactly once. What is the difference between the maximum and minimum number that can be formed?

A 19800

B 41976

C 32976

D None of these

VIEW SOLUTION

Question 78

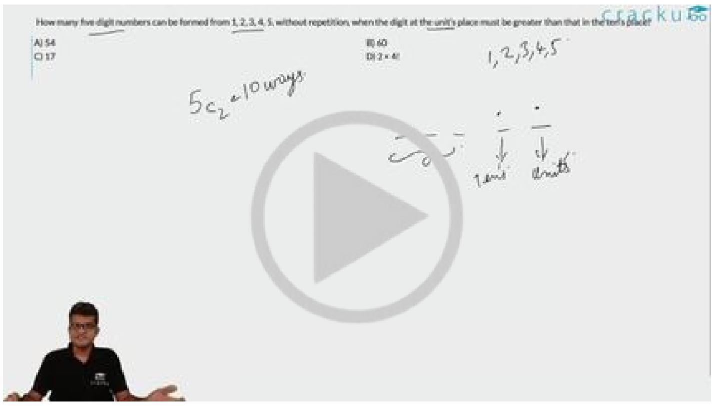
How many five digit numbers can be formed from 1, 2, 3, 4, 5, without repetition, when the digit at the unit's place must be greater than that in the ten's place?

A 54

B 60

C 17

D $2 \times 4!$



VIDEO SOLUTION

Question 79

A certain number, when divided by 899, leaves a remainder 63. Find the remainder when the same number is divided by 29.

A 5

B 4

C 1

D Cannot be determined

VIEW SOLUTION

Question 80

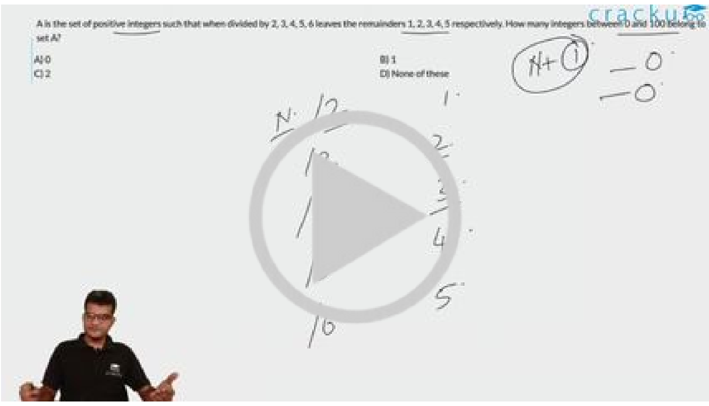
A is the set of positive integers such that when divided by 2, 3, 4, 5, 6 leaves the remainders 1, 2, 3, 4, 5 respectively. How many integers between 0 and 100 belong to set A?

A 0

B 1

C 2

D None of these



VIDEO SOLUTION

Question 81

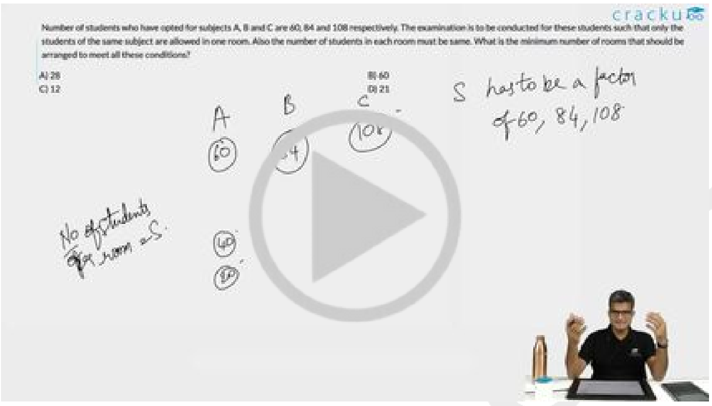
Number of students who have opted for subjects A, B and C are 60, 84 and 108 respectively. The examination is to be conducted for these students such that only the students of the same subject are allowed in one room. Also the number of students in each room must be same. What is the minimum number of rooms that should be arranged to meet all these conditions?

A 28

B 60

C 12

D 21



VIDEO SOLUTION

Question 82

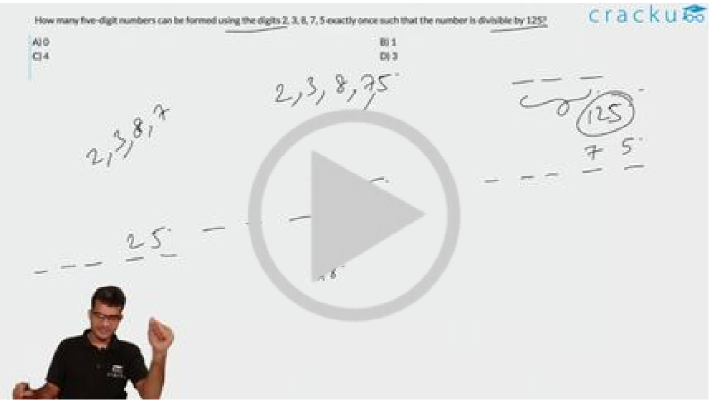
How many five-digit numbers can be formed using the digits 2, 3, 8, 7, 5 exactly once such that the number is divisible by 125?

A 0

B 1

C 4

D 3



VIDEO SOLUTION

Question 83

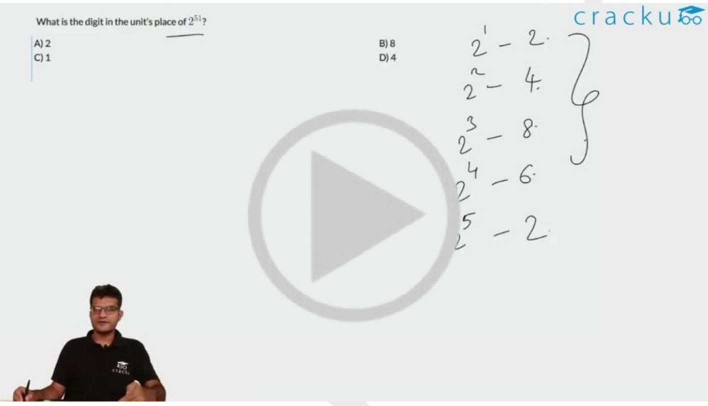
What is the digit in the unit's place of 2^{51}?

A 2

B 8

C 1

D 4



VIDEO SOLUTION

Question 84

A number is formed by writing first 54 natural numbers next to each other as 12345678910111213 ... Find the remainder when this number is divided by 8.

A 1

B 7

C 2

D 0

VIEW SOLUTION

Question 85

Let a, b, c be distinct digits. Consider a two digit number $'ab'$ and a three digit number $'ccb'$, both defined under the usual decimal number system. If $((ab)^2 = ccb)$ and $ccb > 300$ then the value of b is

A 1

B 0

C 5

D 6

VIEW SOLUTION

Question 86

The remainder when 7^{84} is divided by 342 is :

A 0

B 1

C 49

D 341

VIEW SOLUTION

Question 87

If n = 1 + x, where x is the product of four consecutive positive integers, then which of the following is/are true?

A. n is odd

B. n is prime

C. n is a perfect square

A A and C only

B A and B only

C A only

D None of these

VIEW SOLUTION

Question 88

If $n^2 = 123456787654321$, what is n?

A 12344321

B 1235789

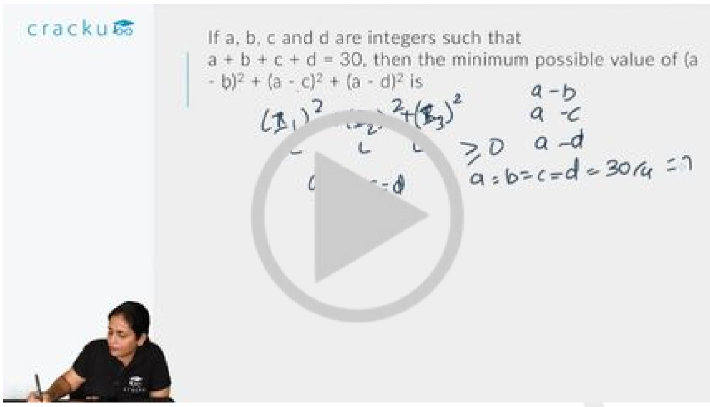
C 11111111

D 1111111

VIEW SOLUTION

Question 89

If a, b, c, and d are integers such that $a + b + c + d = 30$ then the minimum possible value of $(a - b)^2 + (a - c)^2 + (a - d)^2$ is



VIDEO SOLUTION

Question 90

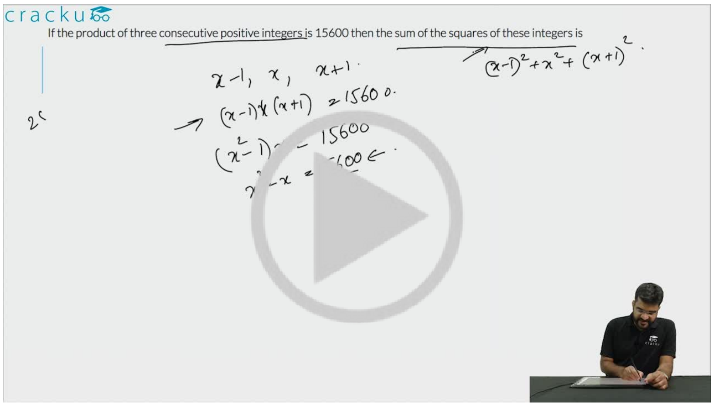
If the product of three consecutive positive integers is 15600 then the sum of the squares of these integers is

A 1777

B 1785

C 1875

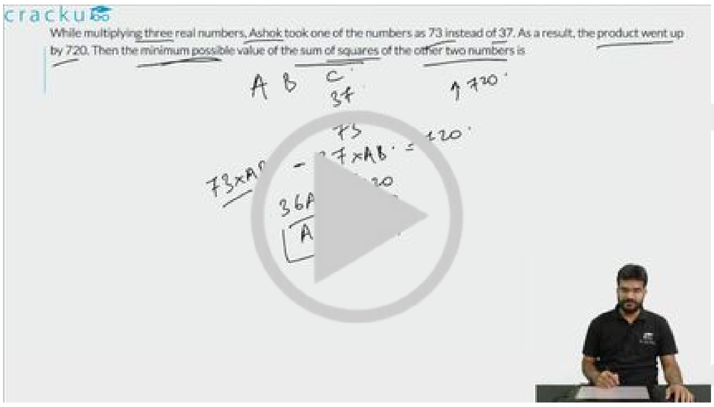
D 1877



VIDEO SOLUTION

Question 91

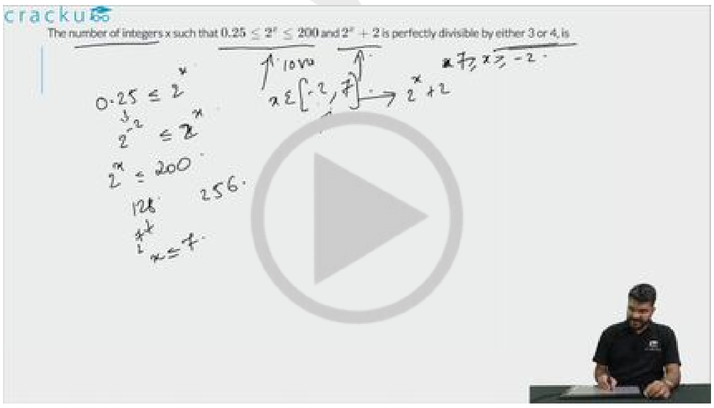
While multiplying three real numbers, Ashok took one of the numbers as 73 instead of 37. As a result, the product went up by 720. Then the minimum possible value of the sum of squares of the other two numbers is



VIDEO SOLUTION

Question 92

The number of integers x such that $0.25 \leq 2^x \leq 200$ and $2^x + 2$ is perfectly divisible by either 3 or 4, is



VIDEO SOLUTION

Question 93

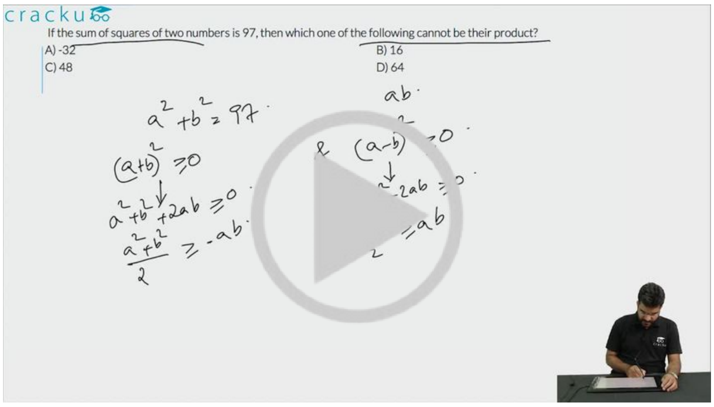
If the sum of squares of two numbers is 97, then which one of the following cannot be their product?

A -32

B 16

C 48

D 64



VIDEO SOLUTION

Question 94

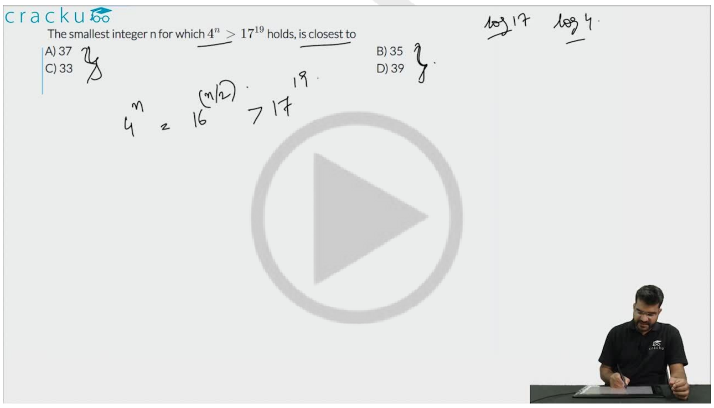
The smallest integer n for which $4^n > 17^{19}$ holds, is closest to

A 37

B 35

C 33

D 39



VIDEO SOLUTION

Question 95

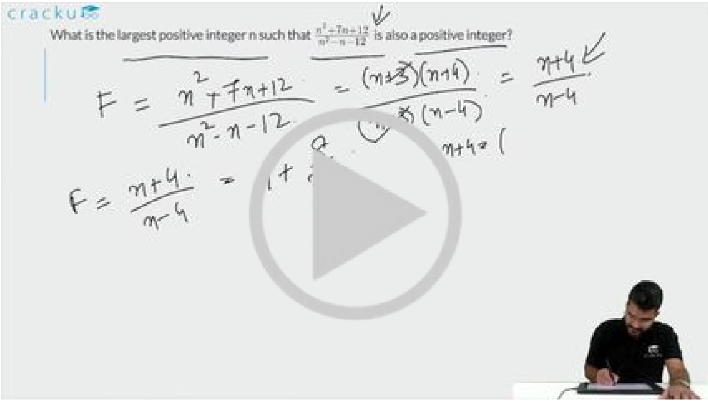
What is the largest positive integer n such that $\frac{n^2+7n+12}{n^2-n-12}$ is also a positive integer?

A 6

B 16

C 8

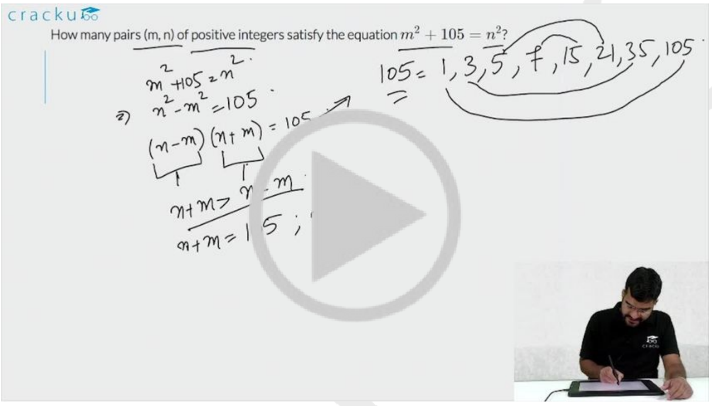
D 12



VIDEO SOLUTION

Question 96

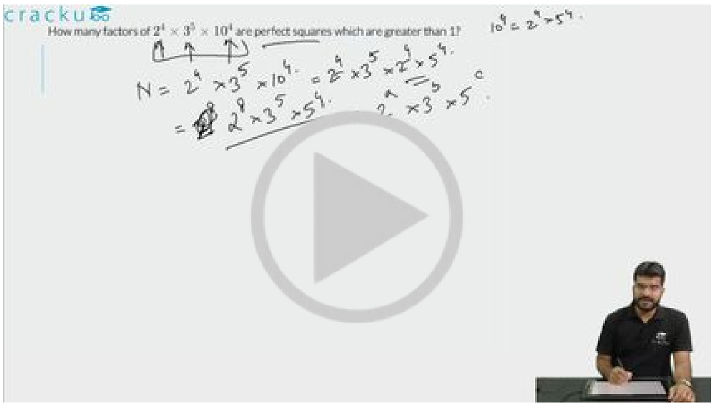
How many pairs (m, n) of positive integers satisfy the equation $m^2 + 105 = n^2$?



VIDEO SOLUTION

Question 97

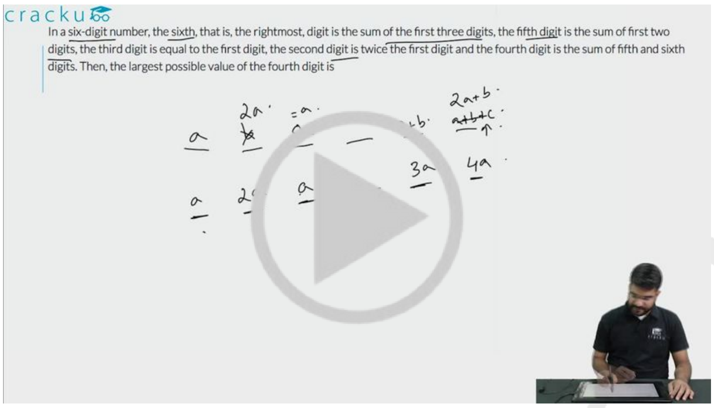
How many factors of $2^4 \times 3^5 \times 10^4$ are perfect squares which are greater than 1?



VIDEO SOLUTION

Question 98

In a six-digit number, the sixth, that is, the rightmost, digit is the sum of the first three digits, the fifth digit is the sum of first two digits, the third digit is equal to the first digit, the second digit is twice the first digit and the fourth digit is the sum of fifth and sixth digits. Then, the largest possible value of the fourth digit is



VIDEO SOLUTION

Question 99

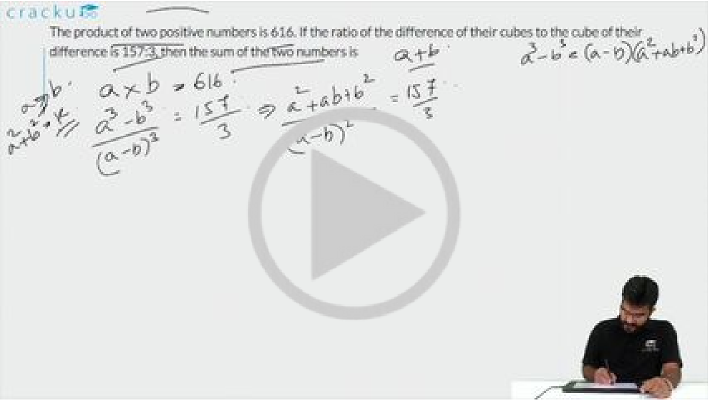
The product of two positive numbers is 616. If the ratio of the difference of their cubes to the cube of their difference is 157:3, then the sum of the two numbers is

A 58

B 85

C 50

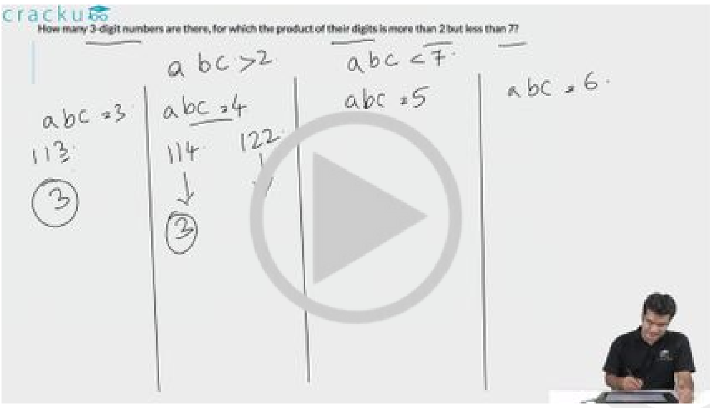
D 95



VIDEO SOLUTION

Question 100

How many 3-digit numbers are there, for which the product of their digits is more than 2 but less than 7?



VIDEO SOLUTION

Question 101

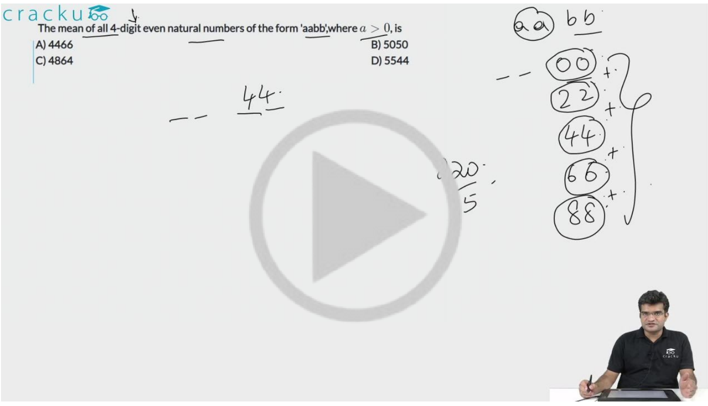
The mean of all 4-digit even natural numbers of the form 'aabb',where $a > 0$, is

A 4466

B 5050

C 4864

D 5544



VIDEO SOLUTION

Question 102

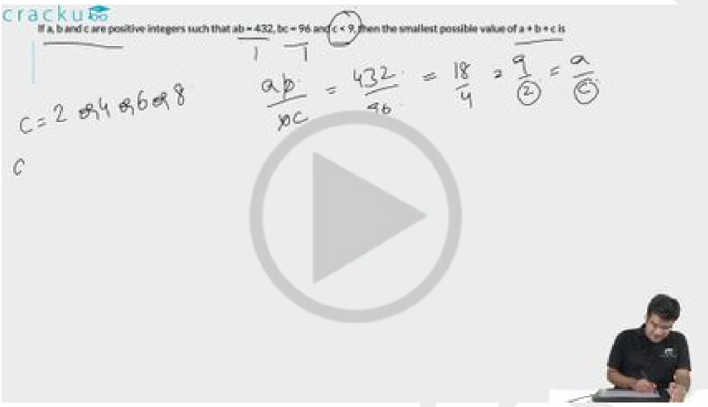
If a, b and c are positive integers such that ab = 432, bc = 96 and c < 9, then the smallest possible value of a + b + c is

A 49

B 56

C 59

D 46



VIDEO SOLUTION

Question 103

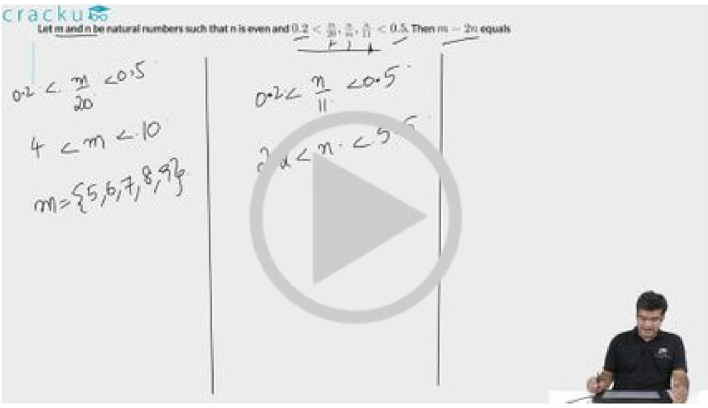
Let m and n be natural numbers such that n is even and $0.2 < \frac{m}{20}, \frac{n}{m}, \frac{n}{11} < 0.5$. Then $m - 2n$ equals

A 3

B 1

C 2

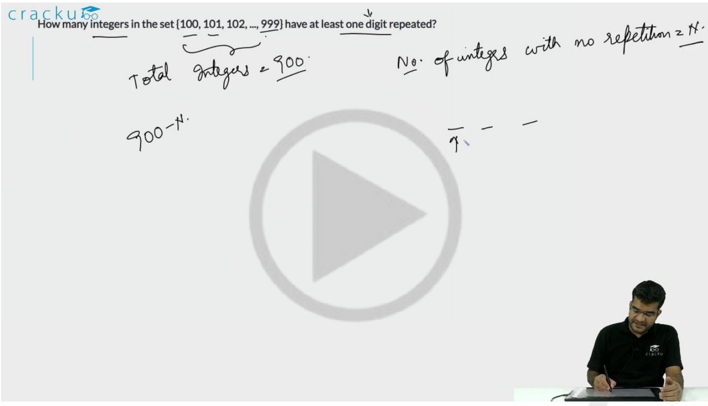
D 4



VIDEO SOLUTION

Question 104

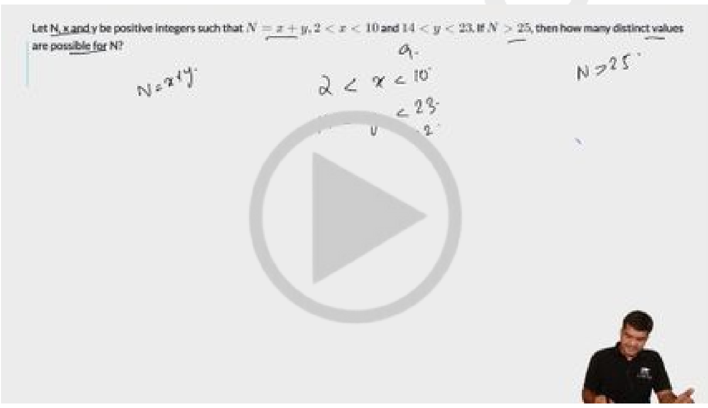
How many integers in the set {100, 101, 102, ..., 999} have at least one digit repeated?



VIDEO SOLUTION

Question 105

Let N, x and y be positive integers such that $N = x + y, 2 < x < 10$ and $14 < y < 23$. If $N > 25$, then how many distinct values are possible for N?



VIDEO SOLUTION

Question 106

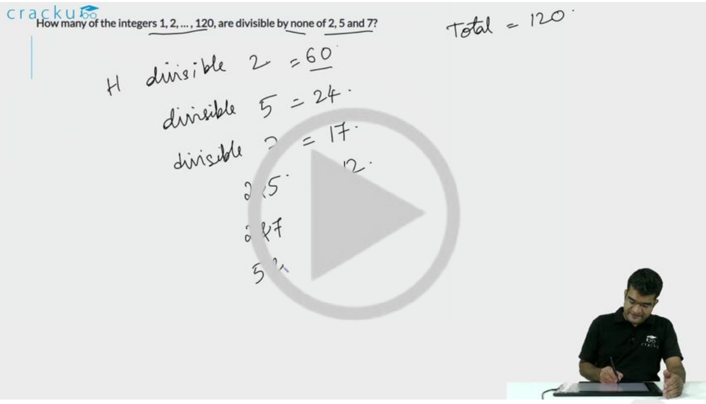
How many of the integers 1, 2, ... , 120, are divisible by none of 2, 5 and 7?

A 42

B 41

C 40

D 43



VIDEO SOLUTION

Question 107

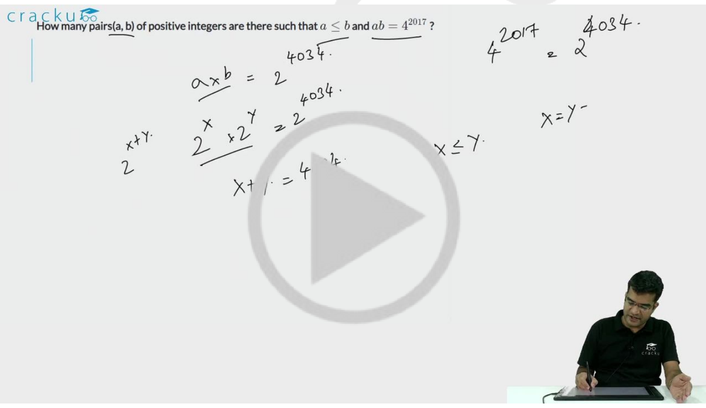
How many pairs(a, b) of positive integers are there such that $a \leq b$ and $ab = 4^{2017}$?

A 2018

B 2019

C 2017

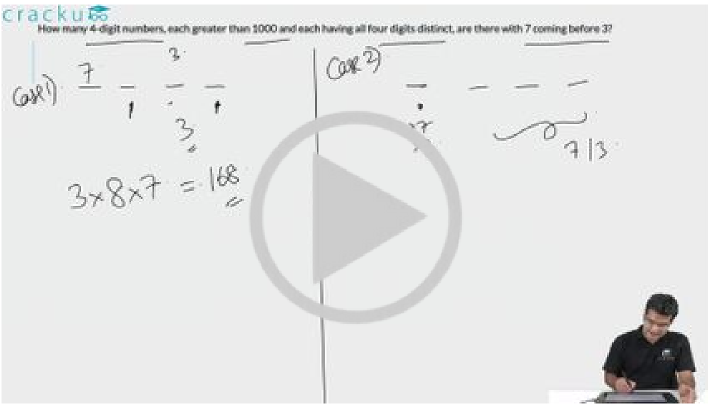
D 2020



VIDEO SOLUTION

Question 108

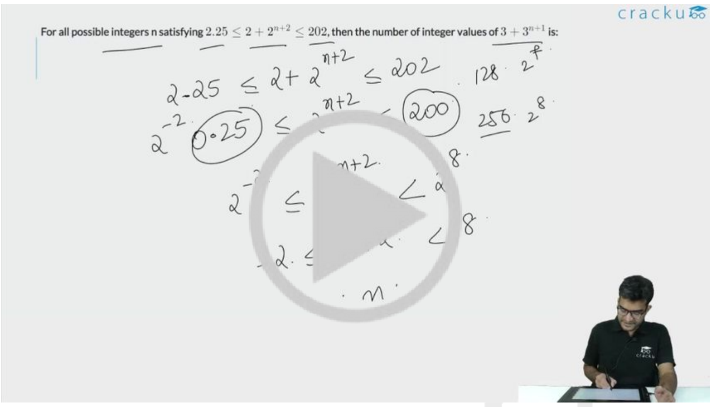
How many 4-digit numbers, each greater than 1000 and each having all four digits distinct, are there with 7 coming before 3?



VIDEO SOLUTION

Question 109

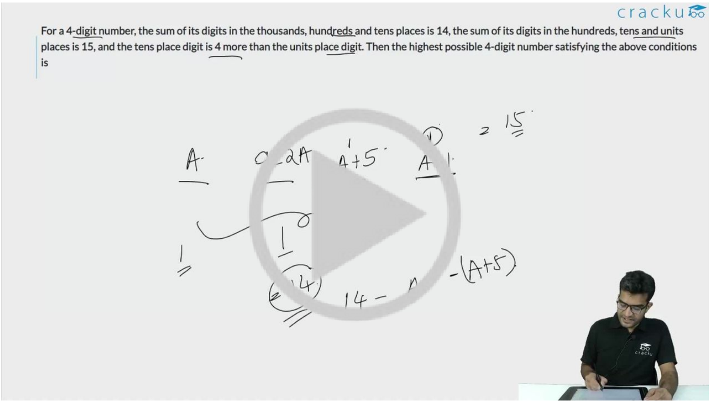
For all possible integers n satisfying $2.25 \leq 2 + 2^{n+2} \leq 202$, then the number of integer values of $3 + 3^{n+1}$ is:



VIDEO SOLUTION

Question 110

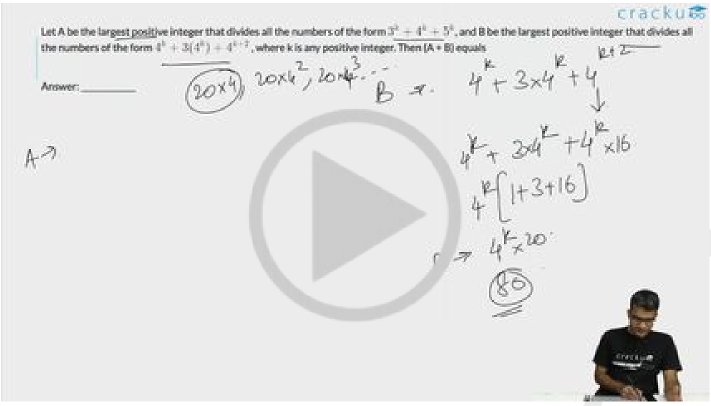
For a 4-digit number, the sum of its digits in the thousands, hundreds and tens places is 14, the sum of its digits in the hundreds, tens and units places is 15, and the tens place digit is 4 more than the units place digit. Then the highest possible 4-digit number satisfying the above conditions is



VIDEO SOLUTION

Question 111

Let A be the largest positive integer that divides all the numbers of the form $3^k + 4^k + 5^k$, and B be the largest positive integer that divides all the numbers of the form $4^k + 3(4^k) + 4^{k+2}$, where k is any positive integer. Then (A + B) equals



VIDEO SOLUTION

Question 112

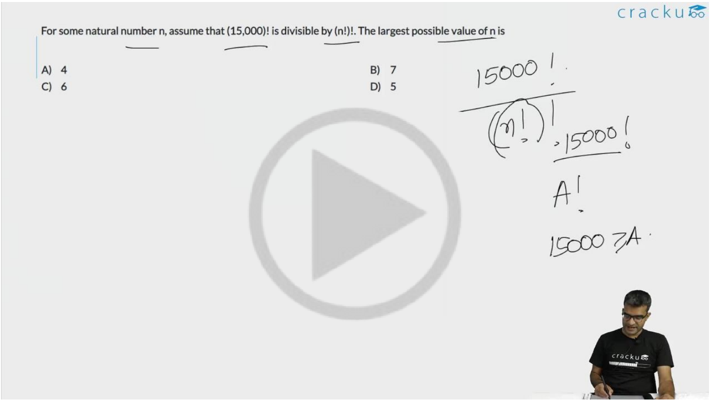
For some natural number n, assume that (15,000)! is divisible by (n!)!. The largest possible value of n is

A 4

B 7

C 6

D 5



VIDEO SOLUTION

Question 113

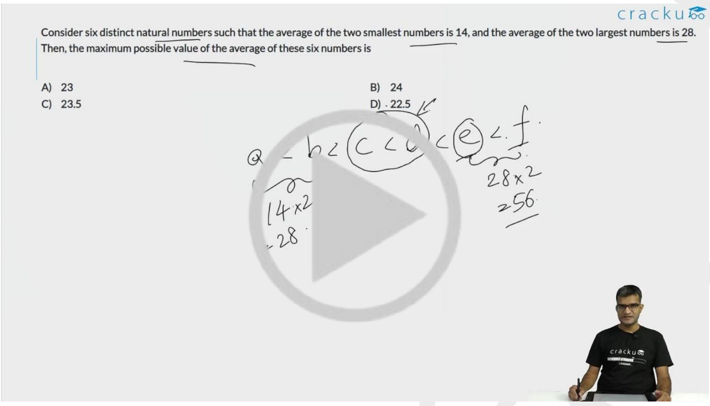
Consider six distinct natural numbers such that the average of the two smallest numbers is 14, and the average of the two largest numbers is 28. Then, the maximum possible value of the average of these six numbers is

A 23

B 24

C 23.5

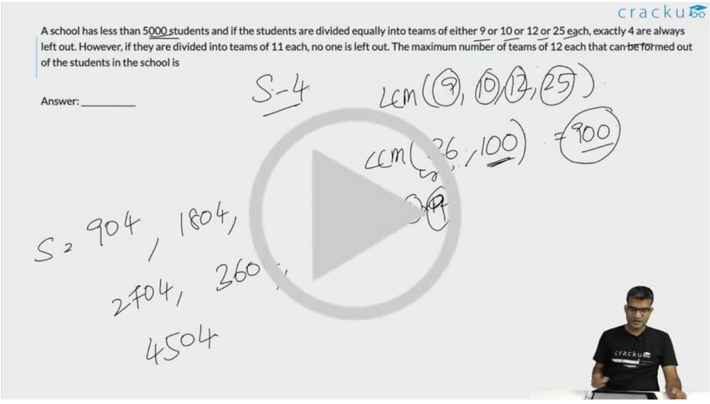
D 22.5



VIDEO SOLUTION

Question 114

A school has less than 5000 students and if the students are divided equally into teams of either 9 or 10 or 12 or 25 each, exactly 4 are always left out. However, if they are divided into teams of 11 each, no one is left out. The maximum number of teams of 12 each that can be formed out of the students in the school is



VIDEO SOLUTION

Question 115

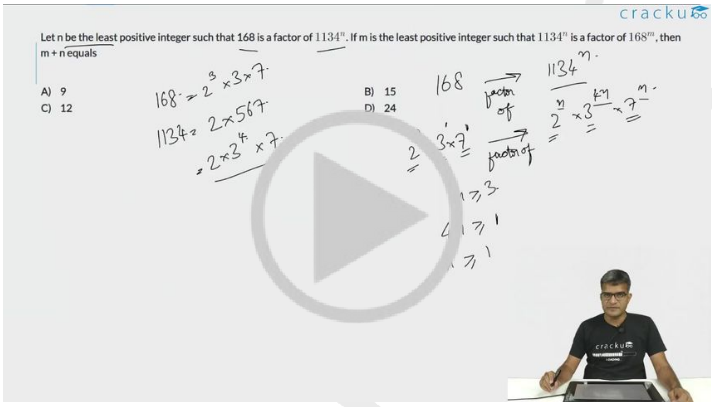
Let n be the least positive integer such that 168 is a factor of 1134^n. If m is the least positive integer such that 1134^n is a factor of 168^m, then m + n equals

A 9

B 15

C 12

D 24



VIDEO SOLUTION

Question 116

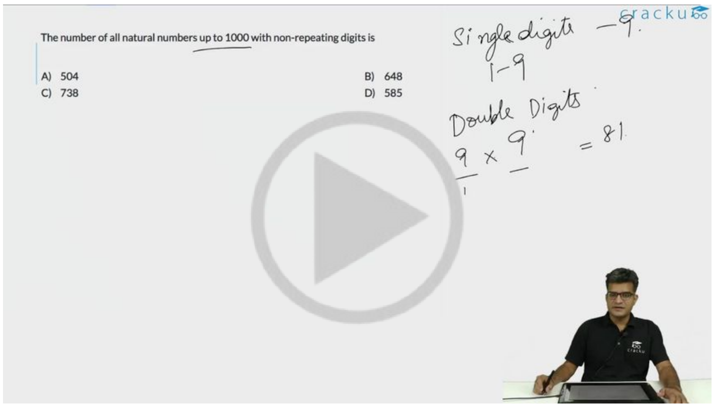
The number of all natural numbers up to 1000 with non-repeating digits is

A 504

B 648

C 738

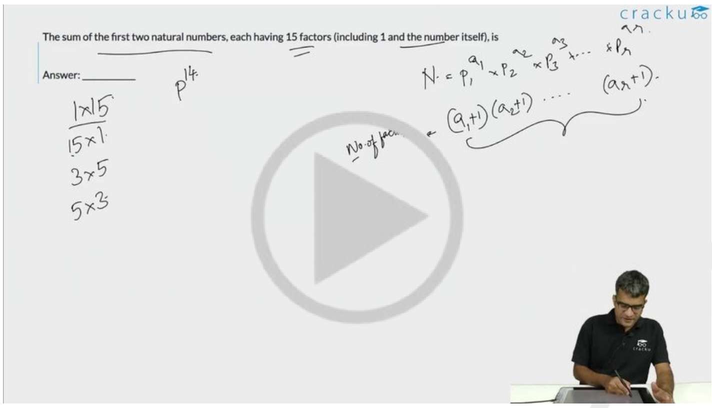
D 585



VIDEO SOLUTION

Question 117

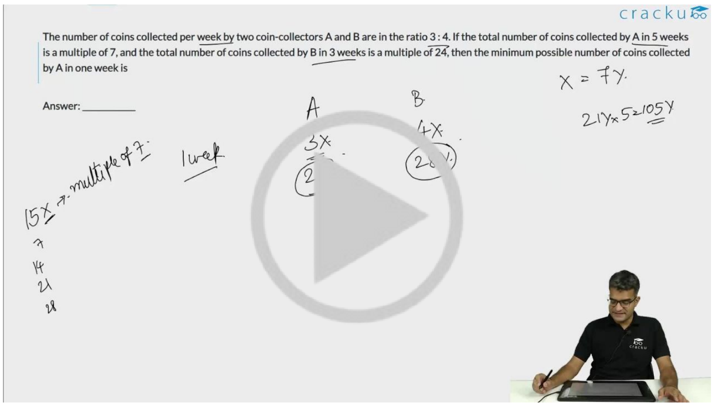
The sum of the first two natural numbers, each having 15 factors (including 1 and the number itself), is



VIDEO SOLUTION

Question 118

The number of coins collected per week by two coin-collectors A and B are in the ratio 3 : 4. If the total number of coins collected by A in 5 weeks is a multiple of 7, and the total number of coins collected by B in 3 weeks is a multiple of 24, then the minimum possible number of coins collected by A in one week is



VIDEO SOLUTION

Question 119

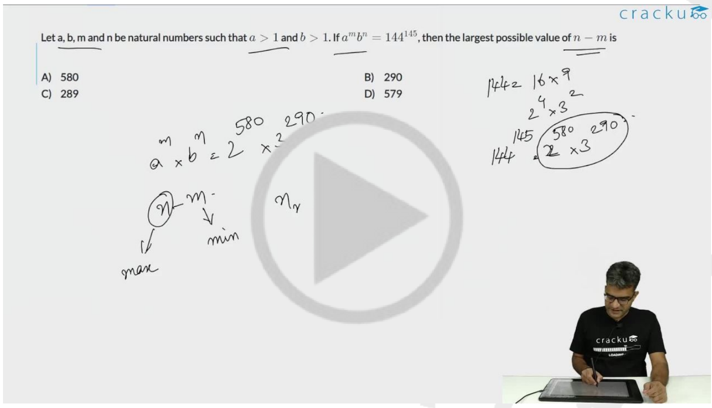
Let a, b, m and n be natural numbers such that $a > 1$ and $b > 1$. If $a^m b^n = 144^{145}$, then the largest possible value of $n - m$ is

A 580

B 290

C 289

D 579



VIDEO SOLUTION

Question 120

For any natural numbers m, n, and k, such that k divides both $m + 2n$ and $3m + 4n$, k must be a common divisor of

A m and n

B 2m and 3n

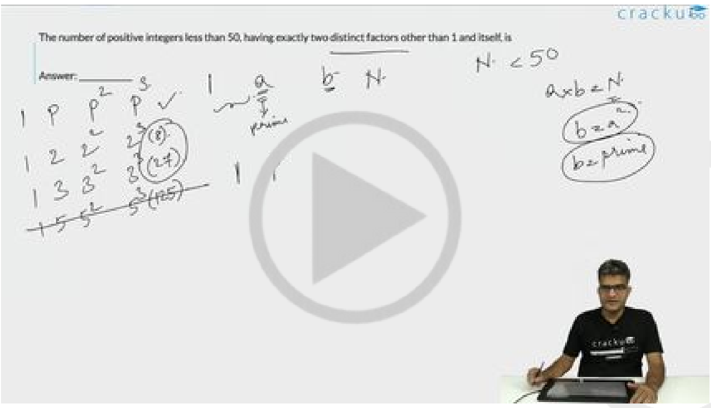
C m and 2n

D 2m and n

VIDEO SOLUTION

Question 121

The number of positive integers less than 50, having exactly two distinct factors other than 1 and itself, is



VIDEO SOLUTION

Question 122

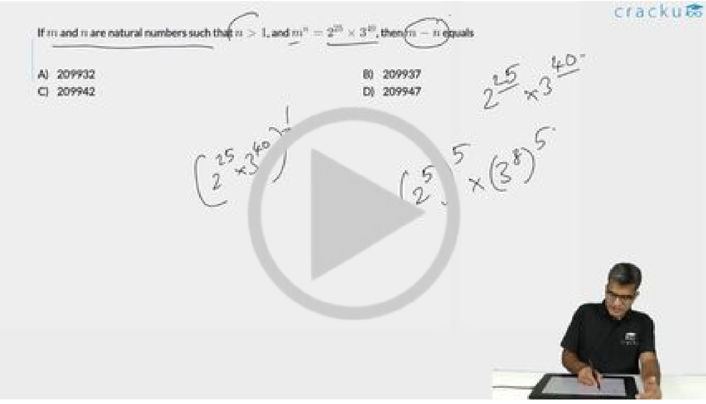
If m and n are natural numbers such that $n > 1$, and $m^n = 2^{25} \times 3^{40}$, then $m - n$ equals

A 209932

B 209937

C 209942

D 209947



VIDEO SOLUTION

Question 123

When 3^{333} is divided by 11, the remainder is

A 5

B 10

C 1

D 6

VIDEO SOLUTION

Question 124

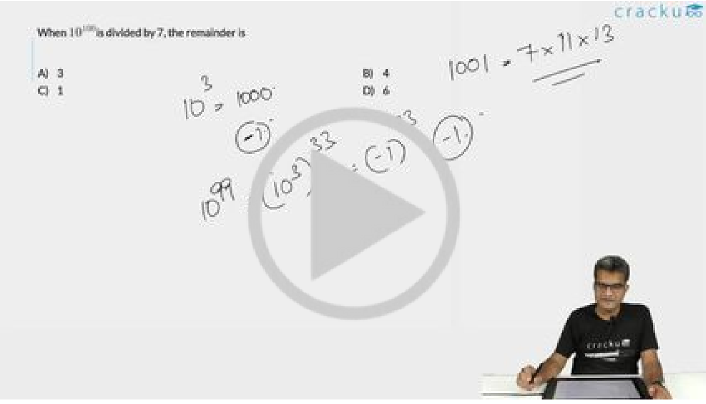
When $10^{100}$is divided by 7, the remainder is

A 3

B 4

C 1

D 6



VIDEO SOLUTION

Question 125

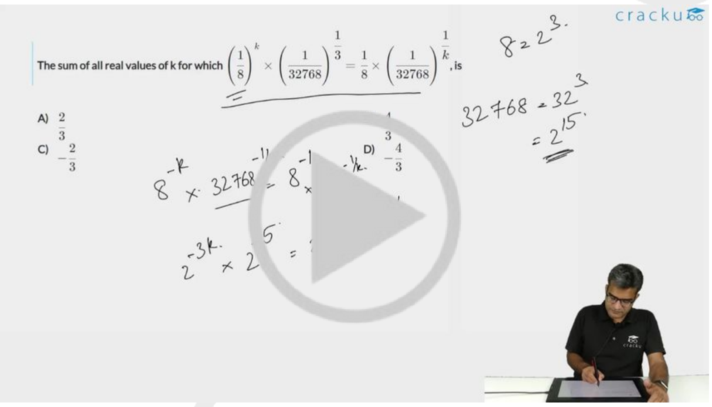
The sum of all real values of k for which $\left(\frac{1}{8}\right)^{k} \times \left(\frac{1}{32768}\right)^{\frac{1}{3}} = \frac{1}{8} \times \left(\frac{1}{32768}\right)^{\frac{1}{k}}$, is

A $\frac{2}{3}$

B $\frac{4}{3}$

C $-\frac{2}{3}$

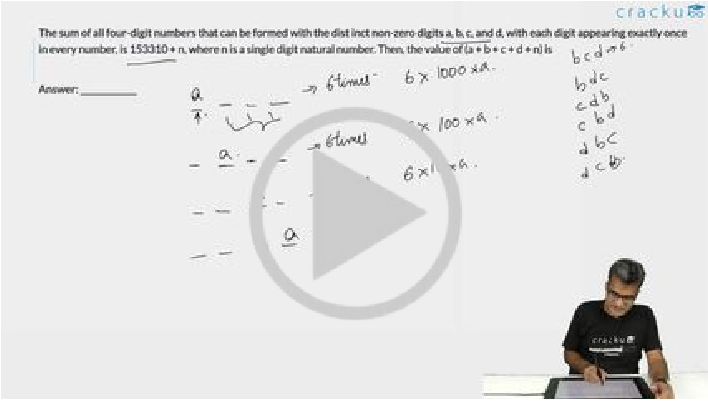
D $-\frac{4}{3}$



VIDEO SOLUTION

Question 126

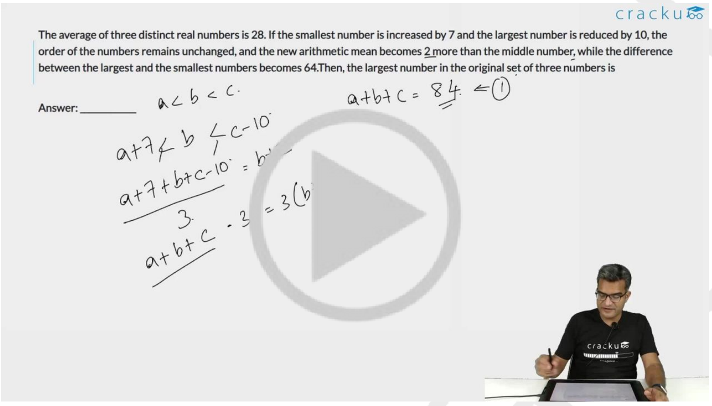
The sum of all four-digit numbers that can be formed with the distinct non-zero digits a, b, c, and d, with each digit appearing exactly once in every number, is 153310 + n, where n is a single digit natural number. Then, the value of (a + b + c + d + n) is



VIDEO SOLUTION

Question 127

The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64.Then, the largest number in the original set of three numbers is



VIDEO SOLUTION

Question 128

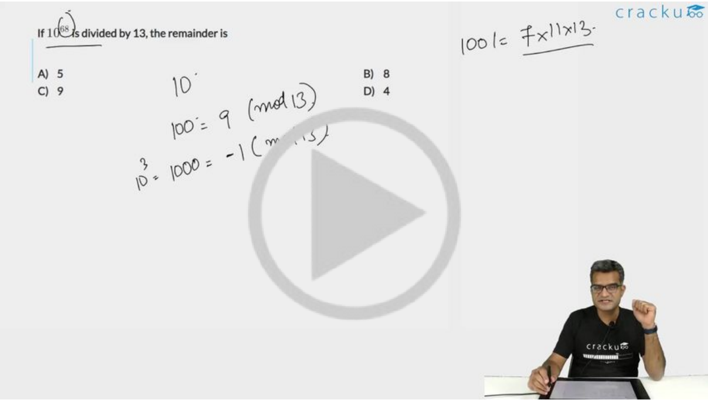
If 10^{68} is divided by 13, the remainder is

A 5

B 8

C 9

D 4



VIDEO SOLUTION

Question 129

The number of all positive integers up to 500 with non-repeating digits is

VIDEO SOLUTION

Instructions

Answer the questions based on the following information. A salesman enters the quantity sold and the price into the computer. Both the numbers are two-digit numbers. But, by mistake, both the numbers were entered with their digits interchanged.Although, the total sales value remained the same, i.e. Rs. 1,148, the total inventory sold got reduced by 54.

Question 130

What is the actual price per piece?

A Rs. 82

B Rs. 41

C Rs. 14

D Rs. 28

VIEW SOLUTION

Question 131

What is the actual quantity sold?

A 28

B 21

C 82

D 14

VIEW SOLUTION

Answers

1.**E**	2.**A**	3.**E**	4.**C**	5.**C**	6.**B**	7.**A**	8.**B**
9.**C**	10.**D**	11.**B**	12.**D**	13.**A**	14.**C**	15.**A**	16.**C**
17.**E**	18.**D**	19.**A**	20.**D**	21.**B**	22.**A**	23.**D**	24.**A**
25.**A**	26.**A**	27.**C**	28.**A**	29.**A**	30.**A**	31.**C**	32.**C**
33.**D**	34.**B**	35.**C**	36.**C**	37.**D**	38.**A**	39.**D**	40.**C**
41.**D**	42.**A**	43.**D**	44.**B**	45.**C**	46.**C**	47.**B**	48.**D**
49.**A**	50.**B**	51.**D**	52.**D**	53.**D**	54.**B**	55.**C**	56.**A**
57.**C**	58.**C**	59.**B**	60.**A**	61.**A**	62.**C**	63.**D**	64.**A**
65.**A**	66.**B**	67.**C**	68.**D**	69.**A**	70.**D**	71.**C**	72.**C**
73.**C**	74.**D**	75.**C**	76.**B**	77.**C**	78.**B**	79.**A**	80.**B**
81.**D**	82.**C**	83.**B**	84.**C**	85.**A**	86.**B**	87.**A**	88.**C**
89.**2**	90.**D**	91.**40**	92.**5**	93.**D**	94.**D**	95.**D**	96.**4**
97.**44**	98.**7**	99.**C**	100.**21**	101.**D**	102.**D**	103.**B**	104.**252**
105.**6**	106.**B**	107.**A**	108.**315**	109.**7**	110.**4195**	111.**82**	112.**B**
113.**D**	114.**150**	115.**B**	116.**C**	117.**468**	118.**42**	119.**D**	120.**C**
121.**15**	122.**D**	123.**A**	124.**B**	125.**C**	126.**31**	127.**70**	128.**C**
129.**378**	130.**C**	131.**C**					

Explanations

1.**E**
1/m + 4/n = 1/12
So, 1/m = 1/12 - 4/n
So, m = 12n/(n-48)
Since m is positive, n should be greater than 48
Also, since n is an odd number, it can take only 49, 51, 53, 55, 57 and 59
If n = 49, 51, 57 then m is an integer, else it is not an integer
So, there are 3 pairs of values for which the equation is satisfied

VIDEO SOLUTION

2.**A**
The number of players in all the teams put together = k * n

The number of players that are common is 1*n = n

So, the number of players in the tournament = kn - n = n(k-1)

VIDEO SOLUTION

3.**E**
Let the number be xxyy
xxyy = 1000x + 100x + 10y +y = 1100x+11y = 11(100x+y)
Since xxyy is a perfect square, and 11 is one of the factors, it should be a multiple of 121
So, xxyy = 121k, where k is also a perfect square.
For k = 4, xxyy is a 3 digit number and for k > 82, xxyy is a five digit number
Between 4 and 82, only for k = 64, the number is of the form xxyy
121*64 = 7744
So, there is only 1 number 7744 which is of the form xxyy and a perfect square.

Alternatively:

The number should be definitely more than 32 and less than 100 as the square is a two digit number.

A number of such form can be written as $(50 \pm a)$ and $100 - a$ where $0 \leq a \leq 100$
So, the square would be of form $(50 \pm a)^2 = 2500 + a^2 \pm 100a$ or $(100 - a)^2$ i.e. $10000 + a^2 + 200a$

In both cases, only a^2 contributes to the tens and ones digit. Among squares from 0 to 25, only 12 square i.e. 144 has repeating tens and ones digit. So, the number can be 38, 62, or 88. Checking these squares only 88 square is in the form of xxyy i.e. 7744.

VIDEO SOLUTION

4.**C**
Let the first operation be (1+40-1) = 40, the second operation be (2+39-1) = 40 and so on

So, after 20 operations, all the numbers are 40. After 10 more operations, all the numbers are 79

Proceeding this way, the last remaining number will be 781

VIDEO SOLUTION

5.**C**
7^4 = 2401 = 2400+1
So, any multiple of 7^4 will always end in 01
Since 2008 is a multiple of 4, 7^{2008} will also end in 01

VIDEO SOLUTION

6.**B**
After the first sale, the remaining quantity would be (x/2)-0.5 and after the second sale, the remaining quantity is 0.25x-0.75

After the last sale, the remaining quantity is 0.125x-(7/8) which will be equal to 0

So 0.125x-(7/8) = 0 => x = 7

VIDEO SOLUTION

7.**A**
Let us assume that three positive consecutive integers are x, x+1, x+2. They are raised to first, second and third powers respectively.

$x^1 + (x+1)^2 + (x+2)^3 = (x + (x+1) + (x+2))^2$

$x^1 + (x+1)^2 + (x+2)^3 = (3x+3)^2$

$x^3 + 7x^2 + 15x + 9 = 9x^2 + 9 + 18x$

After simplifying you get,

$x^3 - 2x^2 - 3x = 0$

=> x=0,3,-1

Since x is a positive integer, it can only be 3.

So, the minimum of the three integers is 3. Option a) is the correct answer.

VIDEO SOLUTION

8. **B**
From the options a, c and d all can possibly occur. Hence option b. Besides, if all people have different number of acquaintances, then first person will have 26 acquaintance, second person will have 25 acquaintance, third person will have 24 and so on till 27 th person will have 0 acquaintance. 0 acquaintance is practically not possible.

VIEW SOLUTION

9. **C**
Between 100 and 200 both included there are 51 even nos. There are 7 even nos which are divisible by 7 and 6 nos which are divisible by 9 and 1 no divisible by both. hence in total 51 - (7+6-1) = 39

There is one more method through which we can find the answer. Since we have to find even numbers, consider the numbers which are divisible by 14, 18 and 126 between 100 and 200. These are 7, 6 and 1 respectively.

VIDEO SOLUTION

10. **D**
Since in all three cases the last digit is 1, the number should give remainder 1 when divided individually by 2,3,5 . So the no. may be 31 or 91 . Now 31 in base 2,3 and 5 give first digit as 1 in all the 3 cases while 91 gives exactly two out of the three cases the leading digit as 1. Hence option D.

VIDEO SOLUTION

11. **B**
positive integers n in the range $12 \leq n \leq 40$ such that the product (n -1)*(n - 2)*...*3*2*1 is not divisible by n, implies that n should be a prime no. So there are 7 prime nos. in given range. Hence option B.

VIDEO SOLUTION

12. **D**
No. of terms in series T , 3+(n-1)*8 = 467 i.e. n=59.

Now S will have atleast have of 59 terms i.e 29 .

Also the sum of 29th term and 30th term is less than 470.

Hence, maximum possible elements in S is 30.

VIDEO SOLUTION

13. **A**

The sum of digits should be 2. The possibilities are 1000000001,1000000010,10000000100,..these 10 cases . Also additional 1 case where 20000000000. Hence option A .

VIEW SOLUTION

14. **C**

The remainder when 15^{23} is divided by 19 equals $(-4)^{23}$
The remainder when 23^{23} is divided by 19 equals 4^{23}
So, the sum of the two equals$(-4)^{23} + (4)^{23} = 0$

VIDEO SOLUTION

15. **A**
a/d = a/b * b/c * c/d = 1/3 * 2 * 1/2 = 1/3

Similarly, b/e and c/f are 3 and 3/8 respectively.

b/e = b/c*c/d*d/e = 3

c/f = c/d*d/e*e/f = 3/8

=> Value of abc/def = 1/3 * 3 * 3/8 = 3/8

VIDEO SOLUTION

16. **C**
Maximum sum of the four numbers <= 384=99+97+95+93
384/10 = 38.4
So, the perfect square is a number less than 38.4
The possibilities are 36, 25, 16 and 9
For the sum to be 360, the numbers can be 87, 89, 91 and 93
The sum of four consecutive odd numbers cannot be 250
For the sum to be 160, the numbers can be 37,39,41 and 43
The sum of 4 consecutive odd numbers cannot be 90
So, from the options, the answer is 41.

VIDEO SOLUTION

17. **E**

The addition of numerator and denominatpr should give a prime no. Only option E gives that.

3 is a factor of 189 and 183 => A and D eliminated

17 is a factor of 187 and 221 => B and C eliminated

181 is prime.

VIEW SOLUTION

18. **D**

$\frac{(30^{65}-29^{65})}{(30^{64}+29^{64})} = ((30-29) * \frac{(30^{64}+30^{63}*29+....+29^{64})}{(30^{64}+29^{64})}$, which is greater than 1 . Hence option D.

VIEW SOLUTION

19. **A**
We know that x = $16^3 + 17^3 + 18^3 + 19^3 = (16^3 + 19^3) + (17^3 + 18^3)$

$= (16+19)(16^2 - 16*19 + 19^2) + (17+18)(17^2 - 17*18 + 18^2)$ = 35 × odd + 35 × odd = 35 × even = 35 × (2k)

=> x = 70k

=> Remainder when divided by 70 is 0.

VIDEO SOLUTION

20. **D**
According to given condiiton we have p = (1 × 1!) + (2 × 2!) + (3 × 3!) + (4 × 4!) + ... + (10 × 10!) . So n × n! = [(n + 1) - 1] × n! = (n + 1)! - n!. So equation becomes p = 2! - 1! + 3! - 2! + 4! - 3! + 5! - 4! +... + 11! - 10!. So p = 11! - 1! = 11! - 1. p + 2 = 11! + 1 .So when it is divided by 11! gives a remainder of 1. Hence, option 4.

VIDEO SOLUTION

21. **B**
Let A = 100x + 10y + z and B = 100z + 10y + x .According to given condition B - A = 99(z - x) As (B - A) is divisible by 7 . So clearly (z - x) should be divisible by 7. z and x can have values 8,1 or 9,2 , such that 8-2=9-2=7 and y can have value from 0 to 9.
So Lowest possible value of A lowest x,y and z which is is 108 and the highest possible value of A is 299.

VIDEO SOLUTION

22. **A**
Rightmost non-zero digit of 30^{2720} is same as rightmost non-zero digit of 3^{272}.

272 is of the form 4k.

All 3^{4k} end in 1.

=> Right most non-zero digit is 1.

VIDEO SOLUTION

23. **D**
Let n can be a 2 digit or a 3 digit number.

First let n be a 2 digit number.
So n = 10x + y and Pn = xy and Sn = x + y
Now, Pn + Sn = n
Therefore, xy + x + y = 10x + y , we have y = 9 .
Hence there are 9 numbers 19, 29,.. ,99, so 9 cases .

Now if n is a 3 digit number.
Let n = 100x + 10y + z
So Pn = xyz and Sn = x + y + z
Now, for Pn + Sn = n ; xyz + x + y + z = 100x + 10y + z ; so. xyz = 99x + 9y .
For above equation there is no value for which the above equation have an integer (single digit) value.

Hence option D.

VIDEO SOLUTION

24. **A**

The no. has all the digits as odd no. and is divisible by 3. So the possibilities are

1113
1119
1131
1137
1155
1173
1179
1191
1197
Hence 9 possibilities .

VIDEO SOLUTION

25. **A**
Take x=3 , z=2 , y=5.

$y(x-z)^2 = 5(3-2)^2 = 5$

Option A gives 5 which is odd.

VIEW SOLUTION

26. **A**
A red light flashes three times per minute and a green light flashes five times in 2 min at regular intervals.
So red light fashes after every 1/3 min and green light flashes every 2/5 min. LCM of both the fractions is 2 min .

Hence they flash together after every 2 min. So in an hour they flash together 30 times .

VIDEO SOLUTION

27. **C**
Each box contains at least 120 and at most 144 oranges.

So boxes may contain 25 different numbers of oranges among 120, 121, 122, ... 144.

Lets start counting.

1st 25 boxes contain different numbers of oranges and this is repeated till 5 sets as 25*5=125.

Now we have accounted for 125 boxes. Still 3 boxes are remaining. These 3 boxes can have any number of oranges from 120 to 144.

Already every number is in 5 boxes. Even if these 3 boxes have different number of oranges, some number of oranges will be in 6 boxes.

Hence the number of boxes containing the same number of oranges is at least 6.

VIDEO SOLUTION

28. **A**

Let the 4 digit no. be xyzw.
According to given conditions we have x+y=z+w --- Eq 1 , x+w=z --- Eq 2,y+w=2x+2z --- Eq 3
Eq 2 - Eq 3 : x-y = -2x-z --- Eq 4
Eq 1+ Eq 4 :2x = -2x+w
4x=w --- Eq 5
Substitute w = 4x inEq2
5x = z
Substitute w = 4x inEq3
y+4x=2x+10x
y = 8x
Now the minimum value x can take is 1 so z =5 and the no. is 1854, which satisfies all the conditions.
Hence option A .

VIDEO SOLUTION

29. **A**
Let the number be X.
From the given information, 53x - 35x = 540 => 18x = 540 => x = 30

So, new product = 35*30 = 1050

VIEW SOLUTION

30. **A**
The product of 44 and 11 in decimal is 484.
If base is x , then 3*x^3+4*x^2+x+4 = 484 .
Hence, the given base system is of number 5.
Now, we have to convert 3111 (in base 5) to decimal number system.

3111 in base 5 equals $1*5^0+1*5^1+1*5^2+3*5^3=1+5+25+375=406$

VIDEO SOLUTION

31. **C**
If the number of pages is 44, the sum will be 44*45/2 = 22*45 = 990
So, the number 10 was added twice

VIEW SOLUTION

32. **C**

We know that a=b^2-b.

Soa^2-a = b(b^3-2b^2-b+2) . = (b - 2)(b - 1)(b)(b + 1)

The above given is a product of 4 consecutive numbers with the lowest number of the product being 2(given b >= 4)

In any set of four consecutive numbers, one of the numbers would be divisible by 3 and there would be two even numbers with the minimum value of the pair being (2,4).

Thus, for any value of b >=4, a^2-4 would be divisible by 3 x 2 x 4 = 24.

Thus, option C is the right choice. Options A and B are definitely wrong as a set of four consecutive numbers need not always include a multiple of 5 eg:(6,7,8,9)

VIDEO SOLUTION

33. **D**
The possible arrangements are 1,multiples of 2 ,remaining. So we have 1+2+4+8+16+32+64+31 = 158. Hence minimum no. of bags required is 8.

VIEW SOLUTION

34. **B**
We have a +c=e so possible summation 6+4=10 or 4+2 = 6.
Also b=2d so possible values 4=2*2 or 10=5*2.
So considering both we have b=10 , d=5, a=4 ,c=2, e=6.
Hence the correct option is B .

VIEW SOLUTION

35. **C**
To be divisible by 4 , last 2 digits of the 5 digit no. should be divisible by 4 . So possibilities are 12,16,32,64,24,36,52,56 which are 8 in number. Remaining 3 digits out of 4 can be selected in $^4C_3$ ways and further can be arranged in 3! ways . So in total = 8*4*6 = 192

VIDEO SOLUTION

36. **C**
Case 1: $a_1 = 0$
So, D equals $0.0a_2 0a_2 0a_2 ...$
So, 100D equals $a_2.0a_2 0a_2 ...$
So, 99D equals a_2

Case 2: $a_2 = 0$
So, D equals $0.a_1 0a_1 0a_1 ...$
So, 100D equals $a_1 0.a_1 0a_1 ....$
So, 99D equals $a_1 0$

So, in both the cases, 99D is an integer. From the given options, only option C satisfies this condition (198=2*99) and hence the correct answer is C.

VIDEO SOLUTION

37. **D**
Number of multiples of 3 between 100 and 200 = 66 - 33 = 33
Number of odd multiples = 16
Number of odd multiples of 21 = 3 (105, 147, 189)
So, the required number = 13

VIDEO SOLUTION

38. **A**

For number of zeroes we must count number of 2 and 5 in prime numbers below 100.

We have just 1 such pair of 2 and 5.

Hence we have only 1 zero.

VIEW SOLUTION

39. **D**

In option d), x-y is even. So, the product of the three terms is even. So, d) cannot be true.

VIEW SOLUTION

40. **C**

The numbers 1421, 1423 and 1425 when divided by 12 give remainder 5, 7 and 9 respectively.

5*7*9 mod 12 = 11 * 9 mod 12 = 99 mod 12 = 3

VIDEO SOLUTION

41. **D**

The difference of the numbers = 34041 - 32506 = 1535

The number that divides both these numbers must be a factor of 1535.

307 is the only 3 digit integer that divides 1535.

VIEW SOLUTION

42. **A**

Since each term is either 1 or -1 . To be 0 we should have even terms and with n=even , no. of terms is even .

VIEW SOLUTION

43. **D**

$55^3 + 17^3 - 72^3$ = $(55 - 72)k + 17^3$. This is divisible by 17

Remainder when 55^3 is divided by 3 = 1

Remainder when 17^3 is divided by 3 = -1

Remainder when 72^3 is divided by 3 = 0

So, $55^3 + 17^3 - 72^3$ is divisible by 3

So, the answer is d) 3 and 17

VIDEO SOLUTION

44. **B**

$(x - y)^2 = x^2 + y^2 - 2xy$

$$0.04 = 0.1 - 2xy => xy = 0.03$$

So, |xy| = 0.03

$$(|x| + |y|)^2 = x^2 + y^2 + 2|xy| = 0.1 + 0.06 = 0.16$$

So, |x|+|y| = 0.4

VIDEO SOLUTION

45. **C**
Quotient of 1982/12 = 165, remainder = 2
Quotient of 165/12 = 13, remainder = 9
Quotient of 13/12 = 1, remainder = 1
Remainder of 1/12 = 1
So, the required number in base 12 = 1192

VIEW SOLUTION

46. **C**
Let x and y be the age of older and younger boy respectively(both single digit). According to given condition we know that $y^2 = x$.

Also Father's age = 10y + x and Mother's age = (10x+y)/2.

Only value which satisfies above equations is x=4 and y=2 .

VIDEO SOLUTION

47. **B**
Consider the options:

24: (Square of sum of digits - the number) = 36 - 24 = 12

54: (Square of sum of digits - the number) = 81 - 54 = 27

34: (Square of sum of digits - the number) = 49 - 34 = 15

45: (Square of sum of digits - the number) = 81 - 45 = 36

So, option b) is the correct answer.

VIDEO SOLUTION

48. **D**
Let x = y+z such that z > y.

We know that $48 * (z-y)^2 = z^2 - y^2$

Solving the above equation, we get z + y = 48

So, option d) is the correct answer.

VIEW SOLUTION

49. **A**

$2^4 = 16 = -1$ (mod 17)
So, $2^{256} = (-1)^{64}$(mod 17)
$= 1$ (mod 17)
Hence, the answer is 1. Option a).

VIDEO SOLUTION

50. **B**

In this problem, the lights are flashed at the intervals 2.5, 4.25 and 5.125 seconds and put off after one second each.

The total duration of intervals of these lights are (2.5+1) = 3.5 s, (4.25+1) = 5.25 s and (5.125+1) = 6.125 s.

We have to find the minimum duration. It would be the LCM of thes three numbers.
Since each word is put after a second. So LCM $[(\frac{5}{2} + 1)(\frac{17}{4} + 1)(\frac{41}{8} + 1)]$ = LCM of numerator / HCF of denominator = 49*3/2 = 73.5. Hence they will glow for full one second after 73.5-1 =72.5 sec.

VIEW SOLUTION

51. **D**

HCF of [(9/2), (27/4), (36/5)] = HCF of numerators / LCM of denominators = 9/20

Total weight = 18.45 lb

So no. of parts = 18.45 / (9/20) = 18.45*20/9 = 41

Hence option d) is the correct answer.

VIDEO SOLUTION

52. **D**

Since after division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively, the number is of form ((((4*4)+1)*3)+2)k = 53K

Let k = 1; the number becomes 53

If it is divided by 84, the remainder is 53.

Option d) is the correct answer.

Alternative Solution.

Consider only for 3 and 4 and the remainders are 2 and 1 respectively.

So 5 is the first number to satisfy both the conditions. The number will be of the form 12k+5. Put different integral values of k to find whether it will leave remainder 5 when divided by 7. So the first number to satisfy such condition is 48x4+5= 53

VIDEO SOLUTION

53. **D**

Consider n=1 we have $7^6 - 6^6$ which is = $(7^3 + 6^3)(7^3 - 6^3)$ = 13 * 127 * 43 which is divisible by all the 3 options.

Option d) is the correct answer.

VIEW SOLUTION

54. **B**
2^{60} or 4^{30} when divided by 5
So according to remainder theorem
remainder will be $(-1)^{30}$ = 1.

VIEW SOLUTION

55. **C**
Take 64 for example.

$\sqrt{64}$ = 8

$\sqrt{8}$ = 2.828

$\sqrt{2.828}$ is close to 1.7

So, we can see that the result is tending towards 1.

VIEW SOLUTION

56. **A**
Putting value of x=0 or 1 and solving all four options,
We will find that only option A satisfies the equation with both values, hence answer will be A.

VIEW SOLUTION

57. **C**
$n^3 - n$ can be written as:
$(n-1)n(n+1)$ (where n is a positive integer)
i.e. product of three consecutive integers.
Hence for any number n=2 or >2 , product will have a factor of 6 in it.
When two numbers are prime in product, then third number will always be divisible by 6
Or product will always have a factor of 3×2 into it.

VIEW SOLUTION

58. **C**
The highest power of 5 in 80! = [80/5] + [$80/5^2$] = 16 + 3 = 19

So, the highest power of 5 which divides 80! exactly = 19

VIDEO SOLUTION

59. **B**
13*62 = 806

31*26 = 806

Hence the answer is option b

VIEW SOLUTION

60. **A**
As 55 and 124 don't have any common factor, and n has to be a minimum integer, Hence, it should be 124 only.
So that given equation won't have a remainder.

VIEW SOLUTION

61. **A**
let's say k+4 = 7m
k = 7m-4
Now for k+2n or 7m+(2n-4) is also multiple of 7.
or 2n-4 should be a multiple of 7
So 2n-4 = 7p
or 2n = 7p+4
For p=2; n=9 (p cannot be 1 as n is an integer)

VIDEO SOLUTION

62. **C**
If 2x+12 is perfectly divisible by x, then 12 must be divisible by x.
Hence, there are six possible values of x : (1,2,3,4,6,12)

VIDEO SOLUTION

63. **D**
Let the Prime number be 6n+1.

So (p^2-1) = 6n(6n+2) = 12n(3n+1)

For any value of n , n(3n+1) will have a factor of 2

Hence given equation will be always be divisible by 24

VIEW SOLUTION

64. **A**
For 203 :
first step = $2\times3^2+0\times3^1+3\times3^0$ = 21
second step = $2\times3^1+1\times3^0$ = 7
So two steps needed to reduce it to 7

VIEW SOLUTION

65. **A**
8+12 =(20 = 2+0) = 2
7+14 =(21 = 2+1) = 3
10+18= (28= 2+8) = 10

VIEW SOLUTION

66. **B**
According to the divisible rule of 9, the sum of all digits should be divisible by 9.
i.e. 55+A+B = 9k
So sum can be either 63 or 72.
For 63, A+B should be 8.
In given options, option B has values of A and B whose sum is 8 and by putting them we are having a number which divisible by both 9 and 8.
Hence answer will be B.

VIDEO SOLUTION

67. **C**
Let's say n=2k+1 where k=1,2,3....
So $n(n^2-1) = 4k(k+1)(2k+1)$
As above expression is already divisible by 4 simultaneously $k(k+1)(2k+1)$ will always be divisible by 6.
Hence complete term will be divisible by 24.

VIEW SOLUTION

68. **D**
Expression can be reduced to 16n + 7 + $\frac{6}{n}$
Now to make above value an integer n can be 1,2,3,6,-1,-2,-3,-6
Hence answer will be D).

VIDEO SOLUTION

69. **A**
let's say number is x.
So $\frac{7x}{8} - \frac{7x}{18} = 770$
or $x = 1584$

VIEW SOLUTION

70. **D**
When PQ = 64

Possible values of P and Q are as follows:

64 and 1 ; 32 and 2 ; 16 and 4 ; 8 and 8

So possible sums are 65,34,20 and 16

Hence answer would be D)

VIEW SOLUTION

71. **C**
Let's say m=5k and n=5t
So m-n = 5(k-t) will be divisible by 5.
$m^2-n^2=25(k^2-t^2)$ will be divisible by 5.
$m+n=5(k+t)$ will be divisible by 5 but not necessarily with 10.

VIEW SOLUTION

72. **C**
Let's say three numbers are (a-2),a,(a+2)

So 3(a-2) = 2(a+2) - 3

a=7 and a+2 = R = 9

VIEW SOLUTION

73. **C**
The values of 1!, 2!, 3!, 4!, 5!, 6! and 7! are 1, 2, 6, 24, 120, 720 and 5040 respectively.

So, the digits must lie from 1 to 6 only to satisfy the conditions.

6 cannot be one of the digits as the at least one digit in the final number is more than 6.

145 = 1! + 4! + 5!

VIDEO SOLUTION

74. **D**
Given expressions can be reduced as follows

A = $\frac{1}{4.000004}$

B = $\frac{1}{6.000003}$

C = $\frac{1}{6.000002}$

Among all of them B is smallest.

VIEW SOLUTION

75. **C**
if n^3 is odd then n will be odd. let's say it is $2k+1$
then n^2 will be = $(4k^2+4k+1)$ which will be odd
Hence answer will be C.

VIDEO SOLUTION

76. **B**

A digit number when squared produces a 3 digit number. This means that the number ranges from [10, 31]. First digit of BE^2 should be unit digit of E^2. But unit digit of E^2 is B. Look at the numbers and the unit digit of their square.

0-0, 1-1, 2-4, 3-9, 4-6, 5-5, 6-6, 7-9, 8-4, 9-1. Only 2-4, 3-9, 4-6, 7-9, 8-4 and 9-1 are kind of pairs we are looking after.But all the pairs except 9-1 produce a number greater than 31. Now, the number we can form from 9-1 is 19 whose square is 361 which satisfies all the condition we are looking for. This is the only such number.

VIDEO SOLUTION

77. **C**

Maximum five digit number which can be formed by using numbers is 43210
And minimum five digit number = 10234
Difference = 43210 - 10234 = 32976

VIEW SOLUTION

78. **B**

Possible numbers with unit's place as 5 = $4 \times 3 \times 2 \times 1 = 24$

Possible numbers with unit's place as 4 and ten's place 3,2,1 = $3 \times 3 \times 2 \times 1 = 18$

Possible numbers with unit's place as 3 and ten's place 2,1 = $2 \times 3 \times 2 \times 1 = 12$

Possible numbers with unit's place as 3 and ten's place 1 = $1 \times 3 \times 2 \times 1 = 6$

Total possible values = 24+18+12+6 = 60

VIDEO SOLUTION

79. **A**

Let's say N is our number
N = (899K + 63) or N = $(29 \times 31$K) + 63
So when it is divided by 29, remainder will be $\frac{63}{29}$ = 5

VIEW SOLUTION

80. **B**

Let the number 'n' belong to the set A.
Hence, the remainder when n is divided by 2 is 1
The remainder when n is divided by 3 is 2
The remainder when n is divided by 4 is 3
The remainder when n is divided by 5 is 4 and
The remainder when n is divided by 6 is 5

So, when (n+1) is divisible by 2,3,4,5 and 6.
Hence, (n+1) is of the form 60k for some natural number k.
And n is of the form 60k-1

Between numbers 0 and 100, only 59 is of the form above and hence the correct answer is 1

VIDEO SOLUTION

81. **D**

As we can see here that total number of students are = 60+84+108 = 252

Now given condition is that in one room only the students of the same subject can be there and the number of rooms should be minimum that means the number of students in a particular room will be maximum.

This Maximum number of students will be HCF (Highest common factor) of 60, 84 and 108 and that will be 12

Hence, number of rooms will be = 252/12 = 21

VIDEO SOLUTION

82. **C**
As we know for a number to be divisible by 125, its last three digits should be divisible by 125
So for a five digit number, with digits 2,3,8,7,5 its last three digits should be 875 and 375
Hence only 4 numbers are possible with its three digits as 875 and 375
I.e. 23875, 32875, 28375, 82375

VIDEO SOLUTION

83. **B**
The last digit of powers of 2 follow a pattern as given below.

The last digit of 2^1 is 2
The last digit of 2^2 is 4
The last digit of 2^3 is 8
The last digit of 2^4 is 6

The last digit of 2^5 is 2
The last digit of 2^6 is 4
The last digit of 2^7 is 8
The last digit of 2^8 is 6

Hence, the last digit of 2^{51} is 8

VIDEO SOLUTION

84. **C**
For a number to be divisible by 8, last 3 digits must be divisible by 8.

Last 3 digits of this number are 354.

354 mod 8 = 2

Hence, 2 is the remainder.

VIEW SOLUTION

85. **A**
$(ab)^2$ = ccb

ccb > 300

The last digit of the number ab must be same as that of the square of ab.

So, b can be 0, 1, 5 or 6.

20^2=400 and 30^2=900 are three digit numbers and greater than 300. But the first 2 digits are not same. Hence, b is not 0.

If b is 5, then the ten's digit of ab's square will be 2 => c = 2. But if c is 2, then ccb is not greater than 300. Hence, b is not 5.

If b is 6, then 26^2 = 576 is the only three digit number that is greater than 300. But, it is not in the form of ccb => b is not 6.

If b is 1, then 21^2=441 satisfies all the given conditions => b is 1.

VIEW SOLUTION

86. **B**
7^3 = 343

7^{84} = $(7^3)^{28}$ = 343^{28}

343^{28} mod 342 = 1^{28} mod 342 = 1

VIEW SOLUTION

87. **A**
Let the four consecutive positive integers be $a, a+1, a+2$ and $a+3$.
Therefore, $n = 1 + a(a+1)(a+2)(a+3)$
Or, $n = 1 + (a^2+3a) * (a^2+3a+2)$
Or, $n = (a^2+3a)^2 + 2*(a^2+3a) + 1 = (a^2+3a+1)^2$
Hence, n is a perfect square and therefore not a prime.

The product of four consecutive positive integers is always even. Hence, n is always odd.
Therefore, from the given statements, only A and C are true.

VIEW SOLUTION

88. **C**
Observe the pattern given below.
$11^2 = 121$
$111^2 = 12321$
$1111^2 = 1234321$ and so on.

So, $11111111^2 = 123456787654321$

VIEW SOLUTION

89. **2**
For the value of given expression to be minimum, the values of a, b, c and d should be as close as possible.
30/4 = 7.5. Since each one of these are integers so values must be 8, 8, 7, 7. On putting these values in the given expression, we get
$(8-8)^2 + (8-7)^2 + (8-7)^2$
=> 1 + 1 = 2

VIDEO SOLUTION

90. **D**
$(x-1)x(x+1) = 15600$
=> $x^3 - x = 15600$
The nearest cube to 15600 is 15625 = 25^3

We can verify that x = 25 satisfies the equation above.
Hence the three numbers are 24, 25, 26. Sum of their squares = 1877

VIDEO SOLUTION

91. **40**
We know that one of the 3 numbers is 37.
Let the product of the other 2 numbers be x.
It has been given that 73x-37x = 720
36x = 720
x = 20

Product of 2 real numbers is 20.
We have to find the minimum possible value of the sum of the squares of the 2 numbers.
Let x=a*b
It has been given that a*b=20

The least possible sum for a given product is obtained when the numbers are as close to each other as possible.
Therefore, when a=b, the value of a and b will be $\sqrt{20}$. ($\sqrt{20}$ is irrational number, but it is a real number)

Sum of the squares of the 2 numbers = 20 + 20 = 40.

Therefore, 40 is the correct answer.

VIDEO SOLUTION

92. **5**
At $x = 0, 2^x = 1$ which is in the given range [0.25, 200]

$2^x + 2$ = 1 + 2 = 3 Which is divisible by 3. Hence, x = 0 is one possible solution.

At $x = 1, 2^x = 2$ which is in the given range [0.25, 200]

$2^x + 2$ = 2 + 2 = 3 Which is divisible by 4. Hence, x = 1 is one possible solution.

At $x = 2, 2^x = 4$ which is in the given range [0.25, 200]

$2^x + 2$ = 4 + 2 = 6 Which is divisible by 3. Hence, x = 2 is one possible solution.

At $x = 3, 2^x = 8$ which is in the given range [0.25, 200]

$2^x + 2$ = 8 + 2 = 3 Which is not divisible by 3 or 4. Hence, x = 3 can't be a solution.

At $x = 4, 2^x = 16$ which is in the given range [0.25, 200]

$2^x + 2$ = 16 + 2 = 18 Which is divisible by 3. Hence, x = 4 is one possible solution.

At $x = 5, 2^x = 32$ which is in the given range [0.25, 200]

$2^x + 2$ = 32 + 2 = 34 Which is not divisible by 3 or 4. Hence, x = 5 can't be a solution.

At $x = 6, 2^x = 64$ which is in the given range [0.25, 200]

$2^x + 2$ = 64 + 2 = 66 Which is divisible by 3. Hence, x = 6 is one possible solution.

At $x = 7, 2^x = 128$ which is in the given range [0.25, 200]

$2^x + 2$ = 128 + 2 = 130 Which is not divisible by 3 or 4. Hence, x = 7 can't be a solution.

At $x = 8, 2^x = 256$ which is not in the given range [0.25, 200]. Hence, x can't take any value greater than 7.

Therefore, all possible values of x = {0,1,2,4,6}. Hence, we can say that 'x' can take 5 different integer values.

VIDEO SOLUTION

93. **D**
Let 'a' and 'b' are those two numbers.

$\Rightarrow a^2 + b^2 = 97$

$\Rightarrow a^2 + b^2 - 2ab = 97 - 2ab$

$\Rightarrow (a - b)^2 = 97 - 2ab$

We know that $(a - b)^2 \geq 0$

$\Rightarrow$ 97-2ab $\geq$ 0

$\Rightarrow$ ab $\leq$ 48.5

Hence, ab $\neq$ 64. Therefore, option D is the correct answer.

VIDEO SOLUTION

94. **D**
$4^n > 17^{19}$

$\Rightarrow 16^{n/2} > 17^{19}$

Therefore, we can say that n/2 > 19

n > 38

Hence, option D is the correct answer.

VIDEO SOLUTION

95. **D**
$\frac{n^2+3n+4n+12}{n^2-4n+3n-12}$

= $\frac{n(n+3)+4(n+3)}{n(n-4)+3(n-4)}$

= $\frac{(n+4)(n+3)}{(n-4)(n+3)}$

= $\frac{(n+4)}{(n-4)}$

= $\frac{(n-4)+8}{(n-4)}$

= $1 + \frac{8}{(n-4)}$ which will be maximum when n-4 =8

n=12

D is the correct answer.

VIDEO SOLUTION

96. **4**
$n^2 - m^2 = 105$

(n-m)(n+m) = 1*105, 3*35, 5*21, 7*15, 15*7, 21*5, 35*3, 105*1.

n-m=1, n+m=105 ==> n=53, m=52

n-m=3, n+m=35 ==> n=19, m=16

n-m=5, n+m=21 ==> n=13, m=8

n-m=7, n+m=15 ==> n=11, m=4

n-m=15, n+m=7 ==> n=11, m=-4

n-m=21, n+m=5 ==> n=13, m=-8

n-m=35, n+m=3 ==> n=19, m=-16

n-m=105, n+m=1 ==> n=53, m=-52

Since only positive integer values of m and n are required. There are 4 possible solutions.

VIDEO SOLUTION

97. **44**

$2^4 \times 3^5 \times 10^4$

=$2^4 \times 3^5 \times 2^4 * 5^4$

=$2^8 \times 3^5 \times 5^4$

For the factor to be a perfect square, the factor should be even power of the number.

In 2^8, the factors which are perfect squares are $2^0, 2^2, 2^4, 2^6, 2^8$ = 5

Similarly, in 3^5, the factors which are perfect squares are $3^0, 3^2, 3^4$ = 3

In 5^4, the factors which are perfect squares are $5^0, 5^2, 5^4$ = 3

Number of perfect squares greater than 1 = 5*3*3-1

=44

VIDEO SOLUTION

98. **7**

Let the six-digit number be ABCDEF

F = A+B+C, E= A+B, C=A, B= 2A, D= E+F.

Therefore D = 2A+2B+C = 2A + 4A + A= 7A.

A cannot be 0 as the number is a 6 digit number.

A cannot be 2 as D would become 2 digit number.

Therefore A is 1 and D is 7.

VIDEO SOLUTION

99. **C**

Assume the numbers are a and b, then ab=616

We have, $\frac{a^3-b^3}{(a-b)^3}$ = $\frac{157}{3}$

=> $3\left(a^3-b^3\right) = 157\left(a^3-b^3+3ab\left(b-a\right)\right)$

=> $154\left(a^3-b^3\right)+3*157*ab\left(b-a\right)$ = 0

=> $154\left(a^3-b^3\right)+3*616*157\left(b-a\right)$ = 0 (ab=616)

=>$a^3-b^3+\left(3\times\ 4\times\ 157\left(b-a\right)\right)$ (154*4=616)

=> $(a-b)\left(a^{2}+b^{2}+ab\right) = 3\times 4\times 157(a-b)$

=> $a^{2}+b^{2}+ab = 3\times 4\times 157$

Adding ab=616 on both sides, we get

$a^{2}+b^{2}+ab+ab= 3\times 4\times 157+616$

=> $(a+b)^{2} = 3\times 4\times 157+616$ = 2500

=> a+b=50

VIDEO SOLUTION

100. **21**

Let the number be 'abc'. Then, $2<a\times b\times c<7$. The product can be 3,4,5,6.

We can obtain each of these as products with the combination 1,1, x where x = 3,4,5,6. Each number can be arranged in 3 ways, and we have 4 such numbers: hence, a total of **12** numbers fulfilling the criteria.

We can factories 4 as 2*2 and the combination 2,2,1 can be used to form **3** more distinct numbers.

We can factorize 6 as 2*3 and the combination 1,2,3 can be used to form **6** additional distinct numbers.

Thus a total of 12 + 3 + 6 = **21** such numbers can be formed.

VIDEO SOLUTION

101. **D**

The four digit even numbers will be of form:

1100, 1122, 1144 ... 1188, 2200, 2222, 2244 ... 9900, 9922, 9944, 9966, 9988

Their sum 'S' will be (1100+1100+22+1100+44+1100+66+1100+88)+(2200+2200+22+2200+44+...)....+(9900+9900+22+9900+44+9900+66+9900+88)

=> S=1100*5+(22+44+66+88)+2200*5+(22+44+66+88)....+9900*5+(22+44+66+88)

=> S=5*1100(1+2+3+...9)+9(22+44+66+88)

=>S=5*1100*9*10/2 + 9*11*20

Total number of numbers are 9*5=45

.'. Mean will be S/45 = 5*1100+44=5544.

Option D

VIDEO SOLUTION

102. **D**

Since $c<9$, we can have the following viable combinations for $b\times c=96$ (given our objective is to minimize the sum):

$48\times 2\,;32\times 3\,;24\times 4\,;16\times 6\,;12\times 8$

Similarly, we can factorize $a\times b=432$ into its factors. On close observation, we notice that 18×24 and 24×4 corresponding to $a\times b$ and $b\times c$ respectively together render us with the least value of the sum of $a+b+c=18+24+4=46$

Hence, Option D is the correct answer.

VIDEO SOLUTION

103. **B**

$0.2 < \frac{n}{11} < 0.5$

=> $2.2<n<5.5$

Since n is an even natural number, the value of n = 4

$0.2 < \frac{m}{20} < 0.5$ => $4< m<10$. Possible values of m = 5,6,7,8,9

Since $0.2 < \frac{n}{m} < 0.5$, the only possible value of m is 9

Hence m-2n = 9-8 = 1

VIDEO SOLUTION

104. **252**

Total number of numbers from 100 to 999 = 900

The number of three digits numbers with unique digits:

_ _ _

The hundredth's place can be filled in 9 ways (Number 0 cannot be selected)

Ten's place can be filled in 9 ways

One's place can be filled in 8 ways

Total number of numbers = 9*9*8 = 648

Number of integers in the set {100, 101, 102, ..., 999} have at least one digit repeated = 900 - 648 = 252

VIDEO SOLUTION

105. **6**

Possible values of x = 3,4,5,6,7,8,9

When x = 3, there is no possible value of y

When x = 4, the possible values of y = 22

When x = 5, the possible values of y=21,22

When x = 6, the possible values of y = 20.21,22

When x = 7, the possible values of y = 19,20,21,22

When x = 8, the possible values of y=18,19,20,21,22

When x = 9, the possible values of y=17,18,19,20,21,22

The unique values of N = 26,27,28,29,30,31

VIDEO SOLUTION

106. **B**

The number of multiples of 2 between 1 and 120 = 60

The number of multiples of 5 between 1 and 120 which are not multiples of 2 = 12

The number of multiples of 7 between 1 and 120 which are not multiples of 2 and 5 = 7

Hence, number of the integers 1, 2, ... , 120, are divisible by none of 2, 5 and 7 = 120 - 60 - 12 - 7 = 41

VIDEO SOLUTION

107. **A**

$ab = 4^{2017} = 2^{4034}$

The total number of factors = 4035.

out of these 4035 factors, we can choose two numbers a,b such that a<b in [4035/2] = 2017.

And since the given number is a perfect square we have one set of two equal factors.

∴ many pairs(a, b) of positive integers are there such that $a \leq b$ and $ab = 4^{2017}$ = 2018.

VIDEO SOLUTION

108. **315**

Here there are two cases possible

Case 1: When 7 is at the left extreme

In that case 3 can occupy any of the three remaining places and the remaining two places can be taken by (0,1,2,4,5,6,8,9)

So total ways 3(8)(7)= 168

Case 2: When 7 is not at the extremes

Here there are 3 cases possible. And the remaining two places can be filled in 7(7) ways.(Remember 0 can't come on the extreme left)

Hence in total 3(7)(7)=147 ways

Total ways 168+147=315 ways

VIDEO SOLUTION

109. **7**

$2.25 \leq 2 + 2^{n+2} \leq 202$

$2.25 - 2 \leq 2 + 2^{n+2} - 2 \leq 202 - 2$

$0.25 \leq 2^{n+2} \leq 200$

$\log_2 0.25 \leq n + 2 \leq \log_2 200$

$-2 \leq n + 2 \leq 7.xx$

$-4 \leq n \leq 7.xx - 2$

$-4 \leq n \leq 5.xx$

Possible integers = -4, -3, -2, -1, 0, 1, 2, 3, 4, 5

If we see the second expression that is provided, i.e

$3 + 3^{n+1}$, it can be implied that n should be at least -1 for this expression to be an integer.

So, n = -1, 0, 1, 2, 3, 4, 5.

Hence, there are a total of 7 values.

VIDEO SOLUTION

110. **4195**

Given the 4 digit number :

Considering the number in thousands digit is a number in the hundredth digit is b, number in tens digit is c, number in the units digit is d.

Let the number be abcd.

Given that a+b+c = 14. (1)

b+c+d = 15. (2)

c = d+4. (3).

In order to find the maximum number which satisfies the condition, we need to have abcd such that a is maximum which is the digit in thousands place in order to maximize the value of the number. b, c, and d are less than 9 each as they are single-digit numbers.

Substituting (3) in (2) we have b+d+4+d = 15, b+2*d = 11. (4)

Subtracting (2) and (1) : (2) - (1) = d = a+1. (5)

Since c cannot be greater than 9 considering c to be the maximum value 9 the value of d is 5.

If d = 5, using d = a+1, a = 4.

Hence the maximum value of a = 4 when c = 9, d = 5.

Substituting b+2*d = 11. b = 1.

The highest four-digit number satisfying the condition is 4195

VIDEO SOLUTION

111. **82**

A is the HCF of $3^k + 4^k + 5^k$ for different values of k.

For k = 1, value is 12

For k = 2, value is 50

For k = 3, value is 216

HCF is 2. Therefore, A = 2

$4^k + 3\left(4^k\right) + 4^{k+2} = 4^k(1+3) + 4^{k+2} = 4^{k+1} + 4^{k+2} = 4^{k+1}(1+4) = 5 \cdot 4^{k+1}$

HCF of the values is when k = 1, i.e. 5*16 = 80

Therefore, B = 80

A + B = 82

VIDEO SOLUTION

112. **B**

To find the largest possible value of n, we need to find the value of n such that n! is less than 15000.

7! = 5040

8! = 40320 > 15000

This implies 15000! is not divisible by 40320!

Therefore, maximum value n can take is 7.

The answer is option B.

VIDEO SOLUTION

113. **D**

Let the six numbers be a, b, c, d, e, f in ascending order

a+b = 28

e+f = 56

If we want to maximise the average then we have to maximise the value of c and d and maximise e and minimise f

e+f = 56

As e and f are distinct natural numbers so possible values are 27 and 29

Therefore c and d will be 25 and 26 respecitively

So average = $\frac{(a+b+c+d+e+f)}{6} = \frac{(28+25+26+56)}{6} = \frac{135}{6} = 22.5$

VIDEO SOLUTION

114. **150**

Since the total number of students, when divided by either 9 or 10 or 12 or 25 each, gives a remainder of 4, the number will be in the form of LCM(9,10,12,25)k + 4 = 900k + 4.

It is given that the value of 900k + 4 is less than 5000.

Also, it is given that 900k + 4 is divided by 11.

It is only possible when k = 2 and total students = 1804.

So, the number of 12 students group = 1800/12 = 150.

VIDEO SOLUTION

115. **B**

Prime Factorising 1134, we get 1134 = $2 \times 3^4 \times 7$ and 168 = $2^3 \times 3 \times 7$

1134^n is a factor of 168 => the factor of 2 should be atleast 3, for 168 to be a factor => n = 3.

Now, 1134^n = $1134^3 = 2^3 \times 3^{12} \times 7^3$ is a factor of $168^m = \left(2^3 \times 3 \times 7\right)^m$ => m = 12 as power of 3 should be atleast 12.

=> So, m + n = 15.

VIDEO SOLUTION

116. **C**

1-digit numbers => We have 1 to 9 => 9

2-digit numbers => x y, we have 9 ways to choose x from 1 to 9 => 9 ways and 9 ways to choose y (0 to 9 except x) => 9*9 = 81

3-digit numbers => x y z, we have 9 ways to choose x, 9 ways to choose y and 8 ways to choose z => 9*9*8 = 648.

Total numbers till 1000 without digits repeated in them is 9 + 81 + 648 = 738.

VIDEO SOLUTION

117. **468**

We know that the number of factors of these two numbers is 15. We know that the factors of 15 are 1, 3, 5, and 15.

The number of factors of N is $(p+1)\cdot(q+1)$(Where, $N = a^p \cdot b^q$, and a, b are prime numbers).

Hence, the value of N will be least when (p+1) and (q+1) are as close as possible and a, and b are the least distinct prime numbers.

Hence, p+1 = 3 => p = 2, and q+1 = 5 => q = 4, and the prime numbers a, and b are 2, and 3, respectively.

Hence, the lowest value of N is $N = 2^4 \times 3^2 = 144$, and the second lowest value of N is $N = 2^2 \times 3^4 = 324$.

Hence, the sum is (144+324) = 468

VIDEO SOLUTION

118. **42**

Given that the number of coins collected per week by two coin-collectors, A and B, are in the ratio 3: 4

Let us assume A collects 3c coins per week and B collects 4c coins per week.

Total number of coins collected by A in 5 weeks = 5*3c = 15c, which should be multiple of 7 => c should be multiple of 7.

Total number of coins collected by B in 3 weeks = 3*4c = 12c, which should be a multiple of 24 => c should be a multiple of 2.

So, the least possible value of c is lcm(2,7) = 14.

Coins sold by A in a week = 3c = 3*14 = 42.

VIDEO SOLUTION

119. **D**

It is given that $a^m \cdot b^n = 144^{145}$, where $a > 1$ and $b > 1$.

144 can be written as $144 = 2^4 \times 3^2$

Hence, $a^m \cdot b^n = 144^{145}$ can be written as $a^m \cdot b^n = \left(2^4 \times 3^2\right)^{145} = 2^{580} \times 3^{290}$

We know that 3^{290} is a natural number, which implies it can be written as a^1, where $a > 1$

Hence, the least possible value of m is 1. Similarly, the largest value of n is 580.

Hence, the largest value of (n-m) is (580-1) = 579

The correct option is D

VIDEO SOLUTION

120. **C**

It is given that k divides m+2n and 3m+4n.

Since k divides (m+2n), it implies k will also divide 3(m+2n). Therefore, k divides 3m+6n.

Similarly, we know that k divides 3m+4n.

We know that if two numbers a, and b both are divisible by c, then their difference (a-b) is also divisible by c.

By the same logic, we can say that {(3m+6n)-(3m+4n)} is divisible by k. Hence, 2n is also divisible by k.

Now, (m+2n) is divisible by k, it implies 2(m+2n) =2m+4n is also divisible by k.

Hence, {(3m+4n)-(2m+4n)} = m is also divisible by k.

Therefore, m, and 2n are also divisible by k.

The correct option is C

VIDEO SOLUTION

121. **15**

Since there are two distinct factors other than 1, and itself, which implies the total number of factors of N is 4.

It can be done in two ways.

First case: $N = p^3$ (where p is a prime number)

Second case: $N = p_1 \times p_2$ (Where p_1, p_2 are the prime numbers)

From case 1, we can see that the numbers which is a cube of prime and less than 50 are 8, and 27 (2 numbers).

From case 2, we will get the numbers in the form (2*3), (2*5), (2*7), (2*11), (2*13), (2*17), (2*19), (2*23), (3*5), (3*7), (3*11), (3*13), (5*7) {(13 numbers)}

Hence, the total number of numbers having two distinct factors is (13+2) = 15.

VIDEO SOLUTION

122. **D**

We must bring the right-hand side in the form so that everything has the same power.

25 has factors 1, 5 and 25

The only common factor 40 and 25 have is 5 (other than 1 of course, which does not work)

So the right-hand side can be rewritten as $(2^5)^5 \times (3^8)^5$

$(32 \times 81 \times 81)^5$

$(209952)^5$

Giving the value of m - n as 209952 - 5 = 209947

Therefore, Option D is the correct answer.

VIDEO SOLUTION

123. **A**

There are multiple ways of solving these sorts of questions. One method is to look for powers of the term in the numerator that leave a remainder of 1 or -1 when divided by the denominator.

Noting down the powers of 3, 3, 9, 27, 81, 243

243 is one such number, 242 is multiple of 11 (11 times 22), hence 243 will leave a remainder of 1 when divided by 11.

243 is 3 raised to power 5; we can rewrite the given term as $\frac{3^{330} \times 3^3}{11}$

The overall remainder will be $\left[\frac{3^{330}}{11}\right]_R \times \left[\frac{3^3}{11}\right]_R$

$\left[\frac{3^{5\times\ 66}}{11}\right]_R \times \left[\frac{3^3}{11}\right]_R$
$\left[\frac{243^{66}}{11}\right]_R \times \left[\frac{3^3}{11}\right]_R$
$1^{66} \times \left[\frac{27}{11}\right]_R$
$1 \times\ 5$
5

Therefore, Option A is the correct answer.

VIDEO SOLUTION

124. **B**
To find the value of $10^{100} mod\,(7)$
When 10 is divided by 7, it leaves a remainder 3, so the above equation can be written as,
$3^{100} mod\,(7)$

Now looking at the cyclicality of powers of 3 when divided by 7,
$3^1 mod\ 7 = 3$

$3^2 mod\ 7 = 2$

$3^3 mod\ 7 = 6$

$3^4 mod\ 7 = 4$

$3^5 mod\ 7 = 5$

$3^6 mod\ 7 = 1$

From this calculation, it is evident that the powers of 3 modulo 7 repeat every 6 steps. This forms a cycle: 3, 2, 6, 4, 5, 1

$3^{100} = \left(3^6\right)^{16} \times\ \left(3^4\right)$

Since $3^6 mod 7 = 1$

We just need to consider $3^4 mod 7$ which equals 4

Hence the answer is 4.

VIDEO SOLUTION

125. **C**
To solve this question, we need to immediately recognise the fact that, $32768 = 8^5$
Substituting this in the above given equation,

$$\left(\frac{1}{8}\right)^k \times \left(\frac{1}{8}\right)^{5\times\ \frac{1}{3}} = \left(\frac{1}{8}\right) \times \left(\frac{1}{8}\right)^{5\times\ \frac{1}{k}}$$

Since the bases are equal, we can equate the powers on either side of the equation,

$k + \frac{5}{3} = 1 + \frac{5}{k}$

$\frac{(3k+5)}{3} = \frac{(k+5)}{k}$

$3k^2 + 5k = 3k + 15$

$3k^2 + 2k - 15 = 0$

Here in the given quadratic equation, the Discriminant is greater than 0, $2^2 - (4)(3)(-15) > 0$
That means both the roots are real, hence we can simply take the sum of the roots of the quadratic equation in k,
Which in a standard quadratic equation of the form $ax^2 + bx + c$ is $-\frac{b}{a}$

Here, the sum of the real values of k is $-\frac{2}{3}$

VIDEO SOLUTION

126. **31**
When gives n distinct numbers, and asked to find the sum of the possible n distinct numbers that can be formed,
We use the formula, $(10^{n-1} + 10^{n-2} + 10^{n-3} + ...) \times ((n-1)!) \times (Sum\ of\ numbers)$

Inserting n=4, $(1000 + 100 + 10 + 1)(3!)(a + b + c + d)$
$(6666)(a + b + c + d)$
We are told that this value equals, $153310 + n$
Since we are told that n is a single digit natural number, the total value cannot be that much greater and 6666 should perfectly divide $153310 + n$

Upon dividing 153310 by 6666 we get the quotient as 22.99
Nearest value being 23, We take 6666x23 giving us the value 153318

Hence the value of n=8 and the value of a+b+c+d=23

Value of a+b+c+d+n=31.

VIDEO SOLUTION

127. **70**
We are given that average of three distinct integers is 28, that means the sum of these three integers is 28x3=84

Let us write $x + y + z = 84$
x, y, z being the three distinct integers in ascending order.

If the smallest number is increased by 7 and the largest number is reduced by 10
$(x + 7) + (y) + (z - 10) = 81$

New arithmetic mean will be $\frac{81}{3} = 27$
And this is said to be 2 more than the middle number, meaning
$27 - 2 = y = 25$

$x + z = 59$

We are given that difference between the largest and the smallest numbers becomes 64,
$(z - 10) - (x + 7) = 64$
$z - x = 81$

Adding the two equations we get, $2z = 140$
$z = 70$

VIDEO SOLUTION

128. **C**

There are multiple ways of solving such questions involving remainders; one easy way is to look for a power of numerator that leaves a remainder of 1 or 01 when divided by the denominator.

In this instance, 1000, when divided by 13, leaves a remainder of -1

We can rewrite the numerator as $\frac{10^{66} \times 100}{13}$

The remainder would be $\left[\frac{10^{66}}{13}\right]_R \times \left[\frac{100}{13}\right]_R$

$(-1)^{22} \times 9$

9

Therefore, Option C is the correct answer.

VIDEO SOLUTION

129. **378**

We need to consider single-digit, two-digit and three-digit numbers

Single digit: There are only nine numbers for a single-digit number (not including zero since we are looking for positive integers only)

Double digits: The ten's digit can be chosen in 9 ways (not including 0), and the unit digit can be chosen in 9 ways (including 0 but excluding the digit used in the ten's place). Hence, a total of 81 numbers.

Three-digit: The hundred's digit can be 1, 2, 3 or 4; hence, there are four options for the hundred's digit; for the ten's digit, there will be 9 options (numbers from 0 to 9 except the one chosen in the hundred's place); for unit's digit there would be 8 options. Hence,e a total of 288 numbers

Giving the total numbers as 288+81+9 = 378

Therefore, 378 is the correct answer.

VIDEO SOLUTION

130. **C**

Total sales value $= 1148 = 4 \cdot 7 \cdot 41$

Let AB be the actual price of the product and CD be the actual quantity.

Since the quantity sold reduced by 54 upon reversing

$CD - DC \ = \ (10 \cdot C + D) - (10 \cdot D + C) = 9 \cdot (C - D)$

wkt, $9 \cdot (C - D) = 54 \ \longrightarrow \ C - D = 6$

Now, wkt, $AB \cdot CD = 1148 \ \ \& \ \ BA \cdot DC = 1148$

The possible values of CD are $93, 82, \& \ 71$

Since only 82 divides 1148, $CD = 82$, Therefore, $AB = 14$

Actual price and Actual quantity are 14 and 82 respectively.

Alternatively,

Since final value $= 1148 = 4 \cdot 7 \cdot 41$ remains the same,

1148 can be represented as product of interchangeable numbers i.e. 41×28 and 82×14

As inventory sold reduced by 54, so, the entry of quantity sold should be 28 and price entry will be 41, and hence, actual price has to be 14 and actual sales volume has to be 82.

VIEW SOLUTION

131. **C**

Total sales value $= 1148 = 4 \cdot 7 \cdot 41$

Let AB be the actual price of the product and CD be the actual quantity.

Since the quantity/inventory increased by 54 upon reversing

$CD - DC \;=\; (10 \cdot C + D) - (10 \cdot D + C) = 9 \cdot (C - D)$

wkt, $9 \cdot (C - D) = 54 \quad \longrightarrow \quad C - D = 6$

Now, wkt, $AB \cdot CD = 1148 \;\;\&\;\; BA \cdot DC = 1148$

The possible values of CD are $93, 82$ & 71

Since only 82 divides 1148, $CD = 82$, Therefore, $AB = 14$

Actual price and Actual quantity are 14 and 82 respectively.

Alternatively,

Since final value $= 1148 = 4 \cdot 7 \cdot 41$ remains the same,

1148 can be represented as product of interchangeable numbers i.e. 41×28 and 82×14

As inventory decreased by 54, so, the quantity sold should be 82 and actual price will be 14.

VIEW SOLUTION